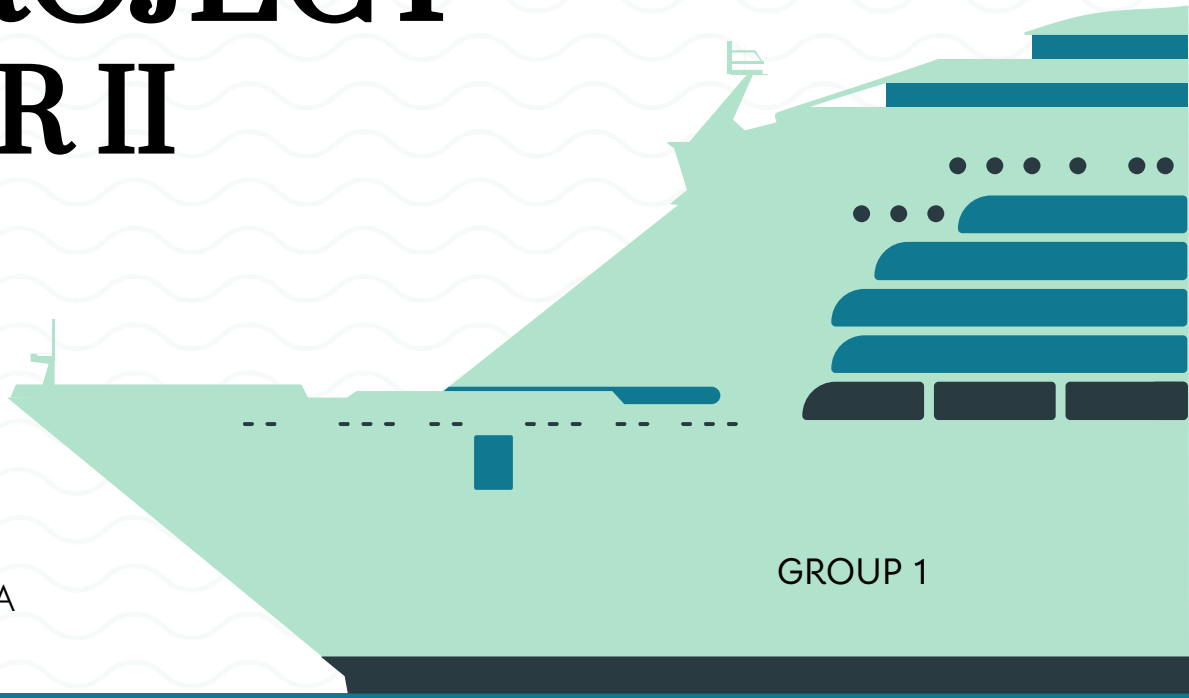
BOAT PROJECT
SEMESTER II

B1 PROMOTION 2029



GROUP 1

Supervisor : Mr MAXENCE MOUANDA

Table of contents

01 MEO Domain

02 MME MAT Domain

05 EEE Domain

03 MME MEC Domain

04 ESE Domain

06 MIA Domain



Internal Organisation of the Group



NYA'ABE NDO Andy Raphaelle

Project Animator and MEO Responsible



NJOUGUEP WANDJI Emmanuelle O

ESE Responsible



ABO William Junior

EEE Responsible



LOUNDOU LOUNDOU Yvon Ezéchiel

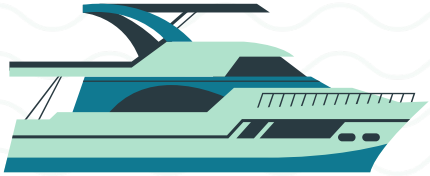
MME Responsible



MAKOSSO Migno JC

MIA Responsible

INTRODUCTION



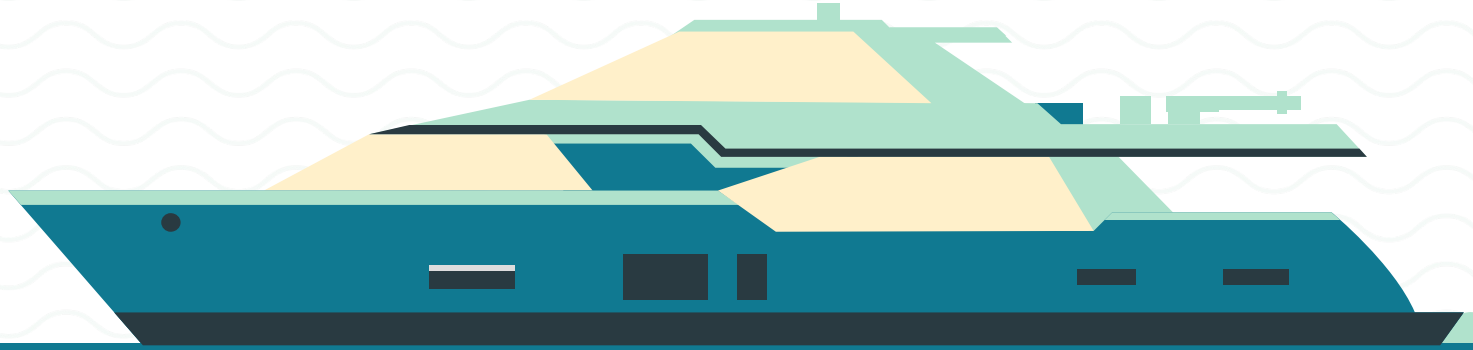


During these months, we were asked to theoretically design a boat hull according to our preferences and well-established standards while respecting the environmental impact.

01

MEO Domain

NYA'ABE NDO ANDY RAPHAELLE



SUMMARY OF THE DOMAIN

- Project presentation
- Project purpose
- Requirement analysis
- Project decomposition
- Project planning (GANTT)
- PERT diagram
- RACI matrix



PROJECT PRESENTATION

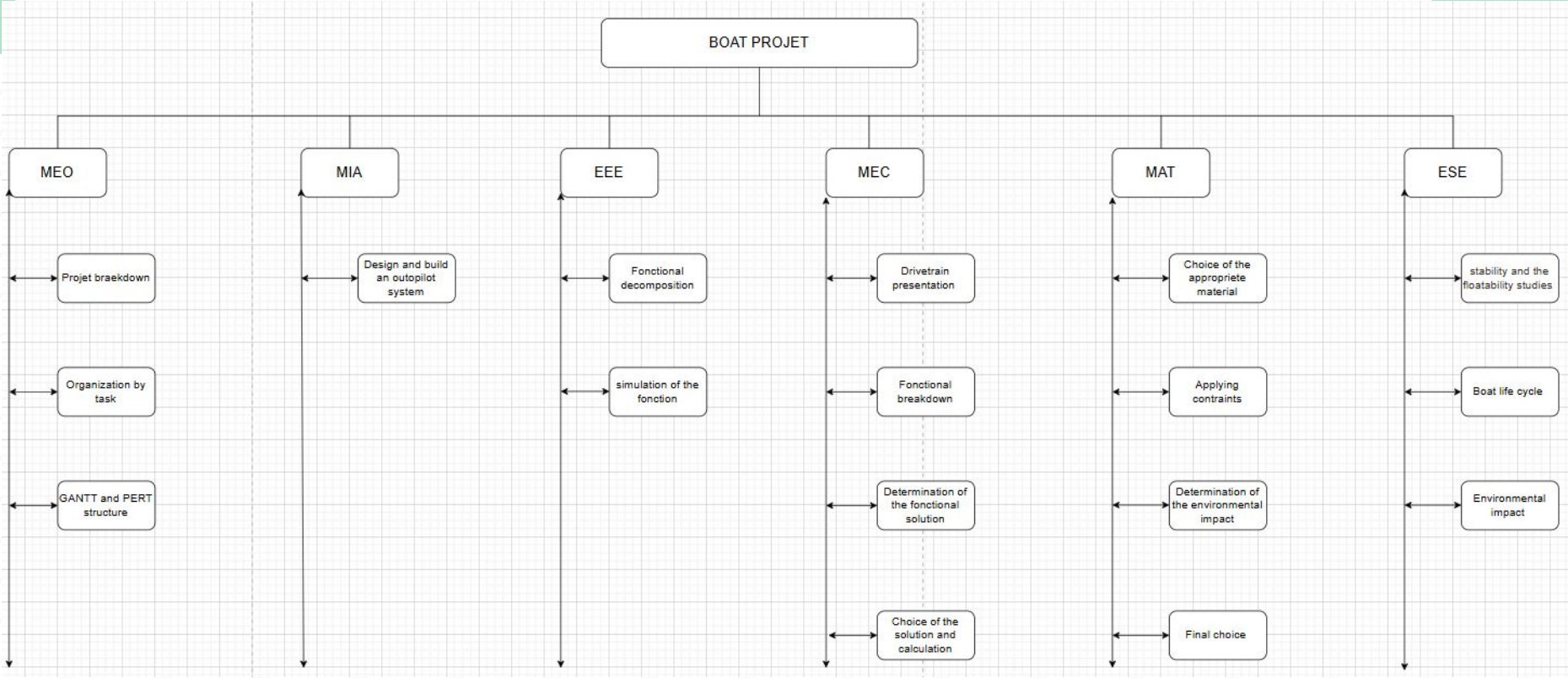
During this first semester, our team was tasked with designing a monohull boat measuring 70 cm in length and including an automatic steering system while minimizing its impact on the environment.

Hours	Distribution
54 Hours	Group work by domain
39 Hours	empowered

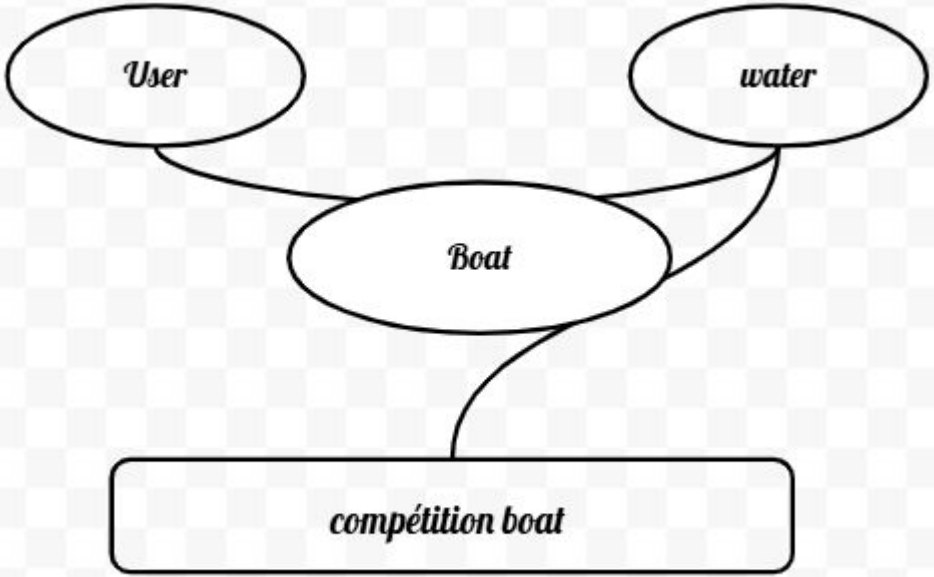
PURPOSE OF THE PROJECT

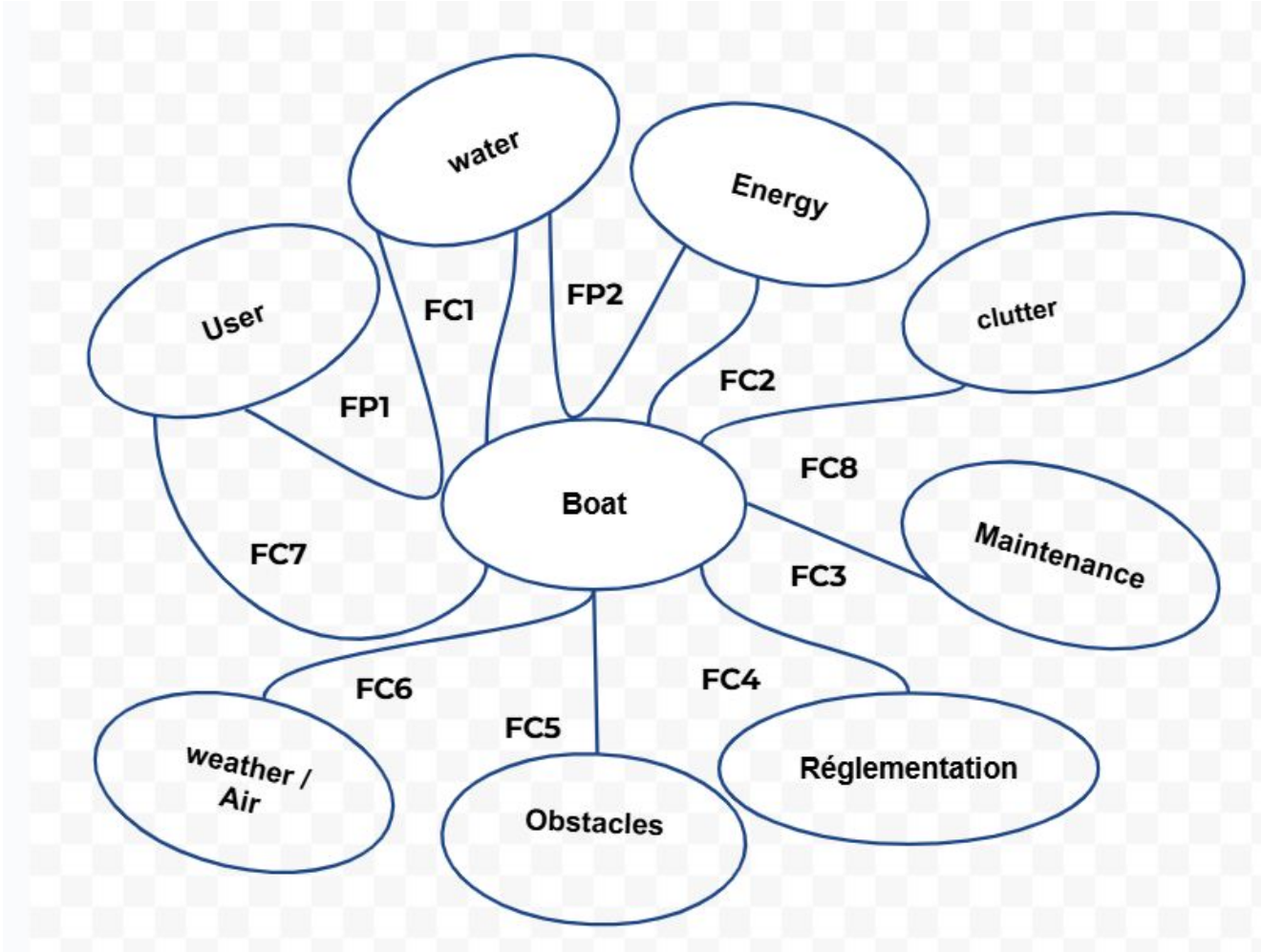
Domaines	Objectives
MEO	Break down the project into tasks and use planning tools to determine how the tasks will be carried out over time
MME	Select and size the power transmission elements and integrate them into the CAD. Choose the material from which the hull will be made and the associated implementation.
EEE	Break down the digital part of the motor control into functional blocks
MIA	Design and production of an automatic piloting system
ESE	Determine the environmental impact of our boat as well as its life cycle

PROJECT DECOMPOSITION



REQUIREMENT ANALYSIS

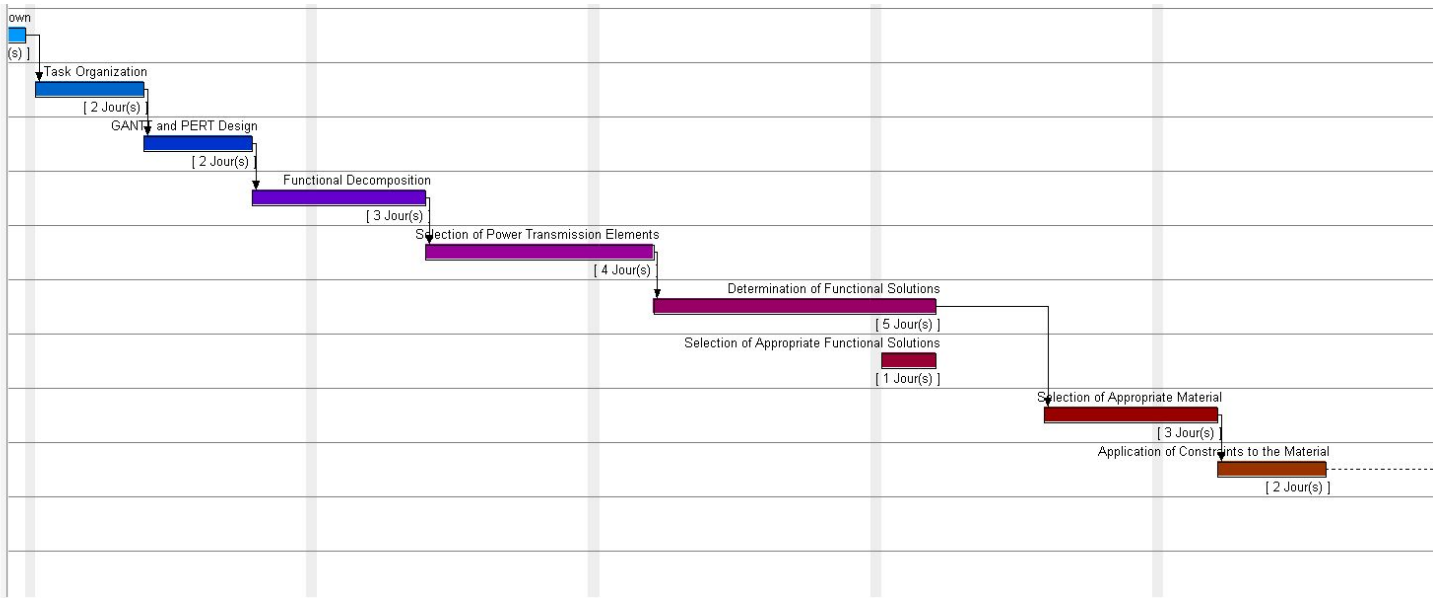



water
Energy
User
FC1
FP2
clutter
FC2
FP1
FC8
Boat
Maintenance
FC7
FC3
FC6
FC4
FC5
weather / Air
Réglementation
Obstacles

FP1: Move on the water following the user's commands
FP2: Allows you to propel the boat
FC1: Float and move on the water without capsizing
FC2: Have an autonomous energy source to power the propulsion and the control system
FC3: Be easy to maintain to prolong its durability
FC4: Comply with regulations relating to remote-controlled devices
FC5: Capable of detecting and avoiding obstacles and unidentified floating objects (OFNI)
FC6: Navigate in varying weather conditions
FC7: Have an easy-to-use control system
FC8: Present reduced dimensions for ease of use

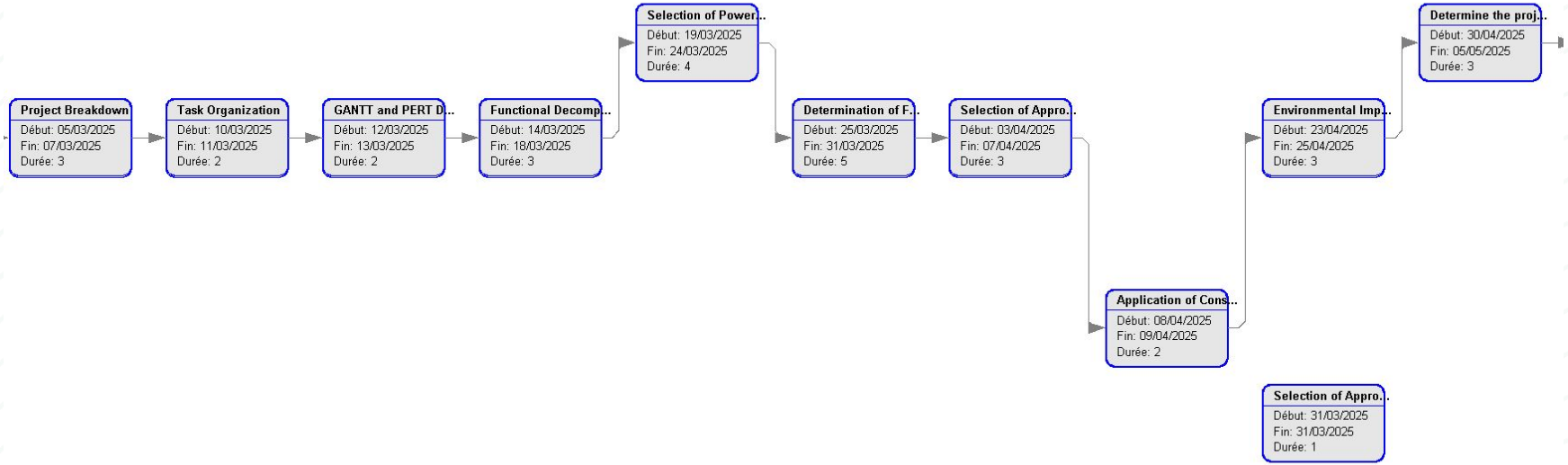
PROJECT PLANNING (GANTT)

Project Breakdown	05/03/2...	07/03/2...
Task Organization	10/03/2...	11/03/2...
GANTT and PERT Design	12/03/2...	13/03/2...
Functional Decomposition	14/03/2...	18/03/2...
Selection of Power Transmis...	19/03/2...	24/03/2...
Determination of Functional ...	25/03/2...	31/03/2...
Selection of Appropriate Fun...	31/03/2...	31/03/2...
Selection of Appropriate Mat...	03/04/2...	07/04/2...
Application of Constraints to ...	08/04/2...	09/04/2...
Environmental Impact	23/04/2...	25/04/2...
Determine the project's carb...	30/04/2...	05/05/2...



PERT DIAGRAM

The PERT chart is a project management tool that allows you to visualize and analyze the different tasks of a project, their dependencies, and the associated deadlines.



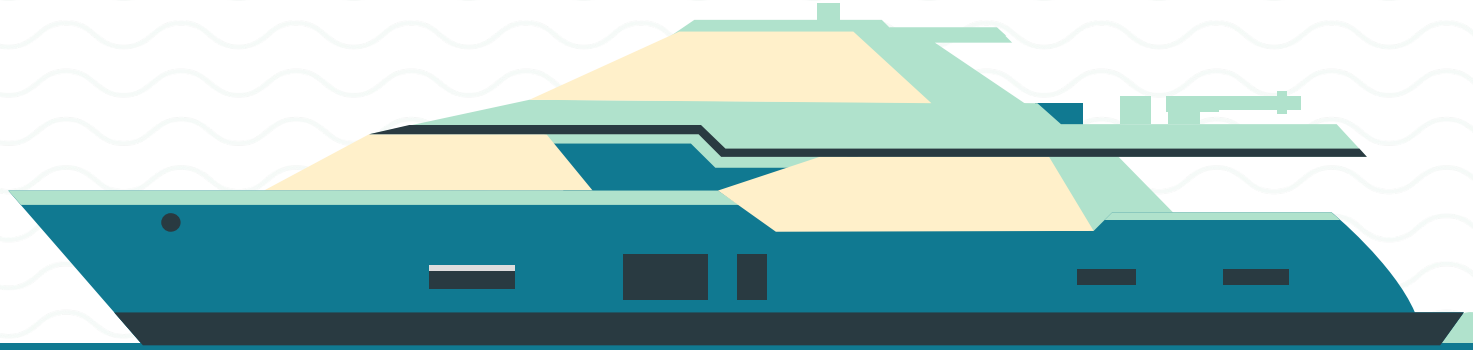
RACI MATRIX

	Administration	Responsable B1	Chef projet	Responsable MEC	Responsable EEE	Responsable MME	Responsable MIA	Responsable ESE	Formateurs	aîné academique
ID Livrable ou tache	Equipe management		Equipe projet						Equipe support	
INITIATION										
Presentaion du projet	A	R	C	C	C	C	C	C	I	I
Guide du projet	A	R	C	C	C	C	C	C	I	I
PLANNIFICATION										
Cahier de charge	I	A	R	R	C	C	C	C	I	I
Etude du projet	I	I	R	R	R	R	R	R	I	I
REALISATION										
Decomposition du projet	I	A	R	R	C	C	C	C	A	I
Organisation par tache	I	A	R	R	C	C	C	C	I	I
Diagramme GANTT et PERT	i	A	R	R	C	C	C	C	I	I
Decomposition fonctionelle	I	A	A	I	R	I	C	I	I	C
Simulation des fonctions	I	I	A	I	R	I	C	I	I	I
Presntation de la chaine de transmission	I	A	I	I	R	I	C	I	I	I
Choix des solutions fonctionnelles	I	I	I	I	R	C	C	I	I	I
Dimensionnement	I	I	I	I	C	R	I	I	I	I
Choix du materiau approprié	I	I	I	C	I	R	I	C	C	I
Application des contraintes	I	I	I	C	C	R	C	I	C	I
Impact environnemental	I	A	I	I	I	R	I	R	A	I
Conception d'un systeme d'autopilotage	I	A	I	I	R	C	C	I	C	I
CONTROLE ET SUIVI										
Envois des livrables	I	A	R	R	R	R	R	R	I	I
Presentation finale	I	A	R	R	R	R	R	R	I	I
CLOTURE										

02

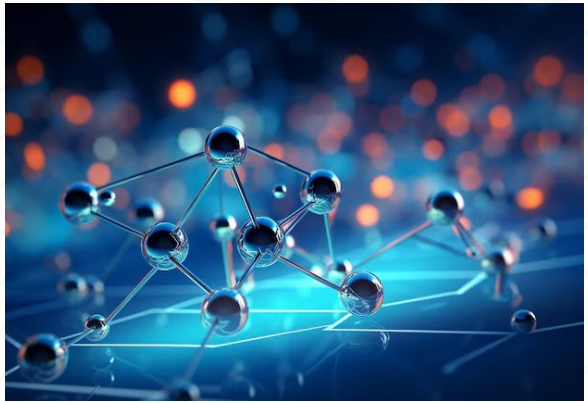
MME MAT Domain

LOUNDOU LOUNDOU YVON EZECHIEL



SUMMARY OF THE DOMAIN

1. Objectives
2. Materials Selection Methodology
   (This section will define the selection criteria for materials)
3. Materials for the Hull and Boat Structure
4. Materials for the Power Transmission System
5. Conclusion



1. Objectives

The objective of this section is to identify and justify the most suitable materials for the construction of our boat. These materials must meet specific requirements such as lightness, mechanical strength, corrosion resistance, ease of manufacturing, compatibility with the aquatic environment, etc. The goal is to ensure optimal performance, increased durability, and cost control, while respecting the technical constraints of the project.

2. Materials Selection Methodology

The hull must meet key technical requirements to ensure the boat's performance and durability in its environment. The main factors for material choice are:

2.1 Material Selection Criteria for the Hull

The boat hull must ensure buoyancy, mechanical resistance to shocks and hydrostatic pressure, and durability in an aquatic environment. The key criteria for selecting hull materials are:

- **Density and specific weight**: A lightweight material is essential to maintain buoyancy and reduce energy consumption.
- **Mechanical strength**: The hull must resist bending, tension, and abrasion caused by waves, impacts with obstacles, or handling.
- **Water and corrosion resistance**: The material must resist freshwater or saltwater exposure without degrading or losing its mechanical properties.
- **Ease of manufacturing**: The material must be easy to shape and assemble, ideally compatible with the planned fabrication processes (molding, lamination, machining).
- **Cost and availability**: The material should be economically feasible and locally available or easy to import.
- **Environmental compatibility**: Durability, recyclability, and ecological impact are increasingly important factors in material selection.

Based on these criteria, candidate materials will be evaluated to determine the best balance between mechanical performance, weight, cost, and durability.

2.2 Material Selection Criteria for Transmission Components

The boat's power transmission system includes several elements (shaft, gears, couplings, stern tube, propeller), each exposed to distinct mechanical, thermal, and environmental stresses. Therefore, material selection must be tailored to the specific constraints of each part.

• Transmission Shaft

Constraints:

- Transfers continuous torque from the motor
- Subject to torsion and sometimes bending
- Partially or fully immersed in water

Selection criteria:

- **High mechanical strength** (shear and fatigue resistance)
- **Corrosion resistance** due to water exposure
- **Good machinability** for bearing and coupling interfaces
- **Sufficient stiffness** to avoid deflection and vibrations
  Cost-effective, especially for longer shafts

• Gears

Constraints:

- Transfer torque via repeated tooth contact
- High friction between teeth → wear risk
- Require precise alignment
- Moderate temperature increase due to friction

Selection criteria:

- **High hardness and wear resistance** to maintain tooth profile
- **Excellent fatigue resistance** for long-term operation
- **Dimensional stability** under load
- **Precision machinability** (gear cutting, shaft fitting)
- **Compatible with lubrication** and surface treatments
- **Heat treatment compatibility** for localized hardening

• Couplings

Constraints:

- Transmit torque between shafts
- May absorb slight misalignments
- Often require periodic disassembly

Selection criteria:

- **Shear and compressive strength**
- **High stiffness** to prevent torque loss
- **Machining accuracy** for tight fitting
- **Wear resistance** for repeated assembly
- **Corrosion resistance** in humid environments
- **Ease of assembly/disassembly** depending on design

• Stern Tube

Constraints:

- Guides and supports the rotating shaft
- Constantly in contact with water
- Sometimes includes a rubber or plastic bearing

Selection criteria:

- **Excellent corrosion resistance** (static underwater component)
- **Compatibility with soft bearings** (no abrasion)
- **Adequate rigidity** to keep shaft aligned
- **Good machinability** for precise mounting
- **Weight is secondary** (non-moving part)
- **Affordable material**

• Propeller

Constraints:

- Direct contact with water
- Subject to thrust forces and dynamic hydrodynamic loads
- Risk of abrasion (sand, debris)

Selection criteria:

- **High corrosion resistance** (especially in saltwater)
- **Strong tensile and bending strength**
- **Shape stability** to maintain blade geometry
- **Smooth surface finish** to reduce drag
- **Moderate weight** to minimize rotational inertia
- **Easy to balance** to avoid vibration

3. Materials for the Hull and Boat Structure

Material	Density / Weight	Mechanical Strength	Corrosion Resistance	Ease of Fabrication	Cost / Availability	Recyclability	Total /35
ABS ASA	4	3	5	4	4	3	23
Marine Wood	3	2	3	3	5	4	20
Fiberglass	4	5	5	3	3	2	22
Carbon Fiber	5	5	5	3	2	2	22

scale: 1 = very poor | 5 = excellent

Recommended choices:

- ABS ASA: offers a good balance between recyclability, performance, and cost

4. Materials for the Power Transmission System

Component	Material	Density (Lightness)	Mechanical Strength	Hardness / Wear	Corrosion Resistance	Machinability / Fit	Cost	Weighted Score
Shaft	Stainless Steel 316L	1	5	4	5	4	3	50
Gear	POM (Delrin)	5	3	4	4	5	4	48
Coupling	Aluminum 7075	4	4	3	3	4	3	42
Stern Tube	Brass	1	3	3	4	3	2	36
Propeller	Glass-Fiber Reinforced Nylon	5	3	3	4	4	3	43

For the transmission chain materials, we conducted a comprehensive analysis using Edupack. This matrix highlights only the main strengths and weaknesses of the selected materials for each component.

5. Conclusion

For the hull, the chosen material is ABS ASA, which offers a good balance between lightness, mechanical strength, corrosion resistance, and recyclability.

For the transmission chain, each component has been matched with the most suitable material:

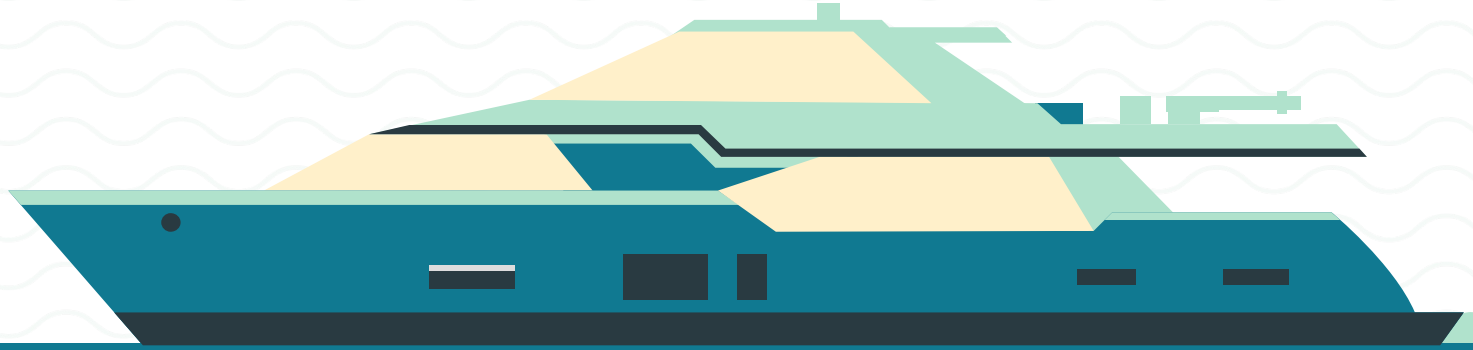
- Shaft in stainless steel 316L
- Gear in POM (Delrin)
- Coupling in aluminum 7075
- Stern tube in brass
- Propeller in glass-fiber reinforced nylon

This overall selection ensures performance, durability, and suitability to the project's constraints.

03

MME MEC Domain

LOUNDOU LOUNDOU YVON EZECHIEL



SUMMARY OF THE DOMAIN

Functional

Mechanical stress calculations

SolidWorks design



Functional diagram

Component Name	Role
DC Motor	Converts electrical energy into mechanical rotation (ω_1, C_1)
Gear	Adjusts rotational speed and/or torque to meet the propeller's needs
Coupling	Mechanically connects two shafts (or a motor and a shaft), compensating for misalignment
Drive Shaft	Transmits rotation from the motor to the propeller
Stern Tube	Holds the shaft axis and ensures watertightness between the hull and water
Propeller	Converts mechanical rotation into thrust in the water

1. Shaft – Torsional Stress

The shaft transmits rotational power from the motor to the propeller. It is primarily subjected to **torsional loading**.

Given:

- Motor power: $P = 15\,\mathrm{W}$
- Rotational speed: $N = 5000\,\mathrm{rpm}$

Transmitted torque:

$$T = \frac{P \times 60}{2\pi N} = \frac{15 \times 60}{2\pi \times 5000} \approx 0.0286\,\mathrm{Nm}$$

Shear stress in a solid circular shaft:

$$\tau = \frac{16T}{\pi d^3}$$

To ensure stress remains below the admissible limit for stainless steel 316L ($\tau_{adm} \approx 200\,\text{MPa}$), we solve for the minimum diameter:

$$d = \left(\frac{16T}{\pi \tau_{adm}}\right)^{1/3} = \left(\frac{16 \times 0.0286}{\pi \times 200 \times 10^6}\right)^{1/3} \approx 3.7\,\text{mm}$$

- **Minimum required shaft diameter: ~4 mm**

---

2. Gears – Contact Stress (Hertzian Theory)

Gears are subjected to **contact pressure** due to rolling and sliding between the teeth.

Simplified assumptions:

- Material: POM (Delrin)
- Admissible contact stress: ~40 MPa
- Stress depends on tooth width, transmitted load and gear diameter (not detailed here due to low power and use of polymer material).

3. Propeller – Bending Stress

Propeller blades experience **bending forces** due to water resistance.

Estimation:

- Load per blade: ~1.5 N
- Blade length: 3 cm
- Estimated section modulus: $Z \approx 1 \times 10^{-9}\,\mathrm{m}^3$
- Bending stress:

$$\sigma = \frac{M}{Z} = \frac{1.5 \times 0.03}{1 \times 10^{-9}} = 45\,\mathrm{MPa}$$

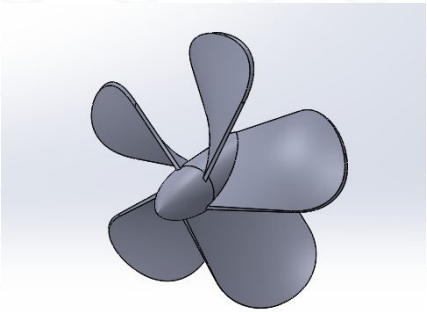
Glass-fiber reinforced nylon has a flexural strength > 70 MPa, which is sufficient.

- **Conclusion:** The propeller material withstands the bending stress under normal operation.

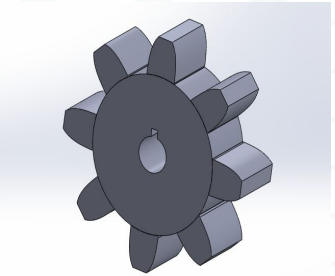
CAO

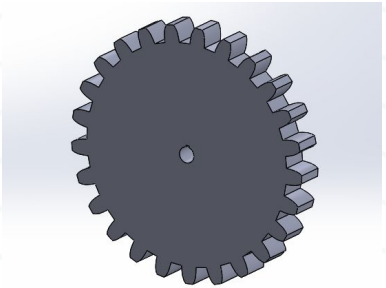


Shaft

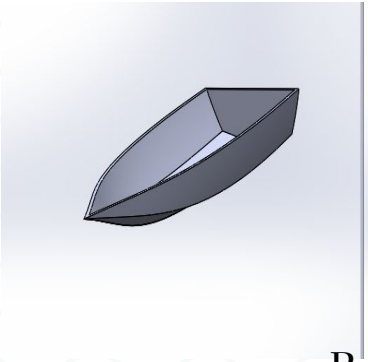


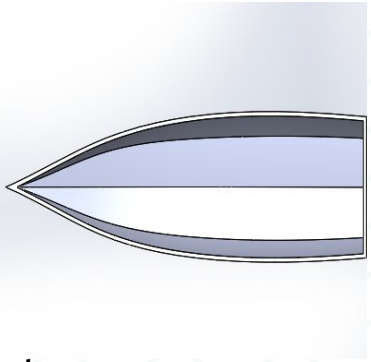
Propeller



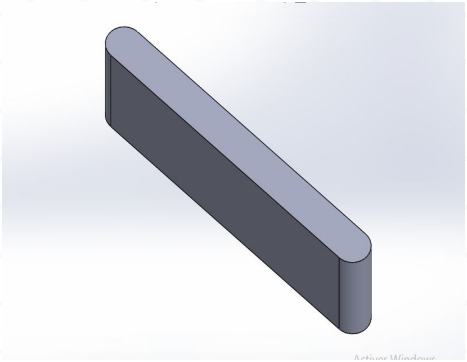


Gear

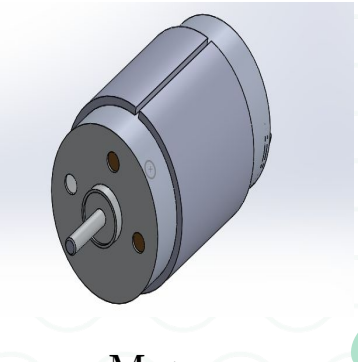




Boat

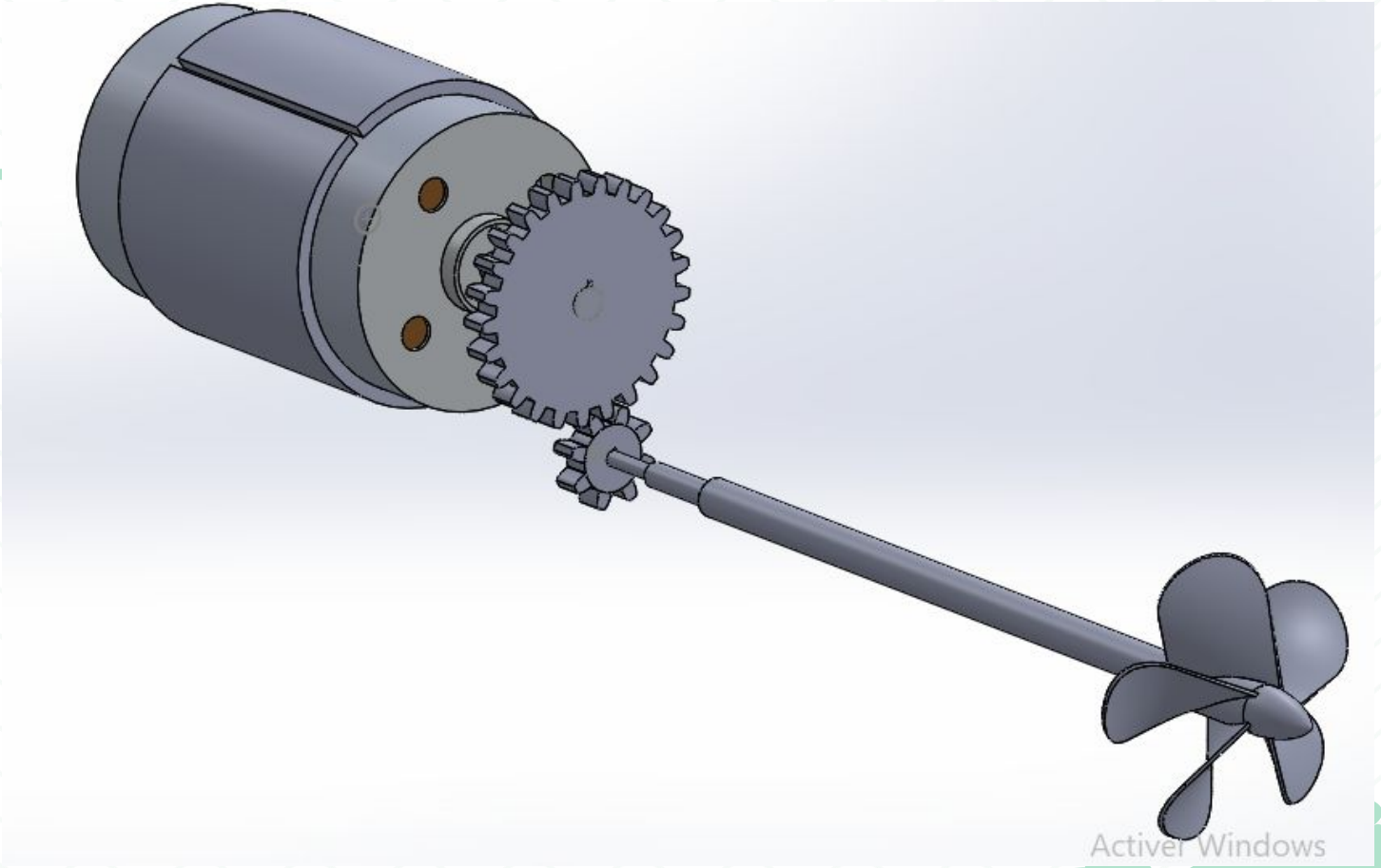


Key



Motor

Power Transmission Assembly

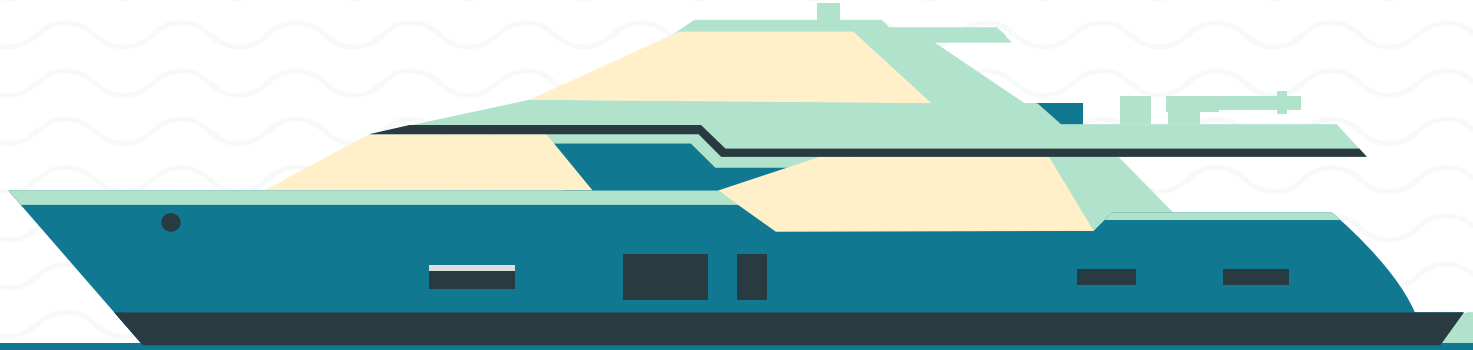


Activer Windows

04

ESE Domain

NJOUGUEP WANDJI EMMANUELLE.O



SUMMARY OF THE DOMAIN

- Objectives of the domain
- Definition of importants terms
- Notion of Buoyancy
- Notion of Stability
- Influencing Factors
- Environmental Impact





OBJECTIFS OF THE SECOND SEMESTER

01

Understand the physical principles applied to buoyancy and stability in floating systems.

02

Apply methods of modeling, calculation, and graphical representation of the forces acting on a floating system.

03

Analyze the technical, environmental, and dynamic factors that influence the behavior of a boat.

04

Learn to integrate electromechanical components in a functional, coherent, and sustainable design approach.

05

Develop environmental awareness by assessing the project's life cycle impact through tools such as LCA and carbon footprint.

I/ DEFINITIONS OF IMPORTANT TERMS

Here are the important terms to know for the ESE domain in the case of our boat project.

1. Archimedes' Force (Buoyant Force):

It is the **upward force** exerted by a fluid (liquid or gas) on a body **partially or completely submerged** in it.

- This force is equal to the **weight of the fluid displaced** by the object.
- It acts **vertically upward** through the center of buoyancy.

2. Apparent Weight:

This is the **effective weight** of an object **as measured when it is submerged in a fluid**.

- It is calculated as:
  Apparent Weight= True Weight − Buoyant Force

- So, an object in water seems **lighter** than when in air because of the upward buoyant force.

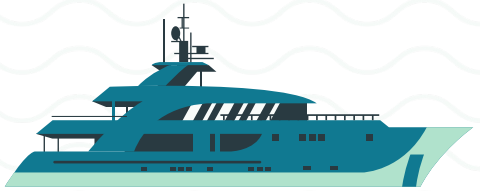
3. Heeling:

Heeling refers to the **temporary tilt** of a ship to one side due to **external forces** such as wind, waves, or turning.

- It is usually **reversible** and the ship returns to upright position when the force is removed.
- Common in sailing vessels when wind pushes against the sails.

4. Careen:

To careen a ship means to **tilt it deliberately** to one side (onto its side or keel), typically to perform **maintenance or repairs** below the waterline.



- This is often done in shallow water or on a sloping shore.
- Different from heeling, which is due to external, dynamic forces.

5. Archimedes' Principle (with Hypothesis):

Statement: *"Any object, wholly or partially immersed in a fluid, experiences an upward force (buoyant force) equal to the weight of the fluid it displaces."*

Hypothesis/Assumptions:

- The fluid is **at rest** (hydrostatic condition).
- The object is **immersed in the fluid** (partially or fully).
- The fluid is **incompressible** and **homogeneous**.
- Gravity is uniform throughout the fluid.

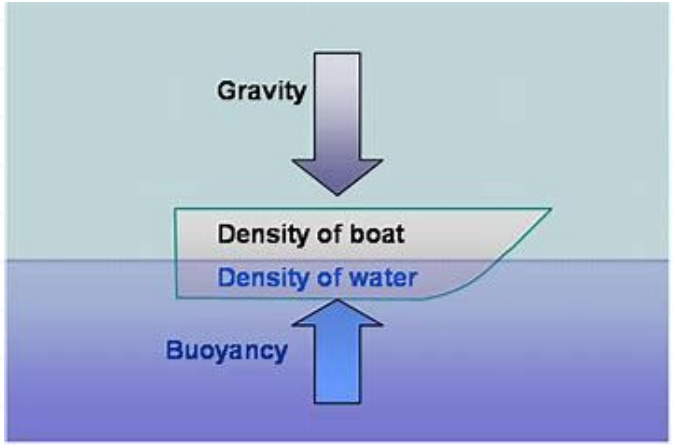
II/ NOTION OF BUOYANCY

Buoyancy is the ability of a boat to stay afloat due to Archimedes' principle. This force is equal to the weight of the volume of water displaced by the boat.

1. Archimedes' Principle :

The buoyant force is given by this formula : Fa = .g.V

p = density of freshwater (Kg/m3)
g = gravitational acceleration (m²/s)
V = volume of water displaced by the hull (m3)



2. Flotation conditions of a boat :

- f Fa > p : The boat will rise further out of the water.
- If Fa < p : The boat will sink.
- If Fa = p : The boat is in flotation equilibrium.

3. Boat weight calculation :

So, The boat weight (P) = m .g = 2,2 . 9,81 21.9 N

Components	Weight
DC Motor	157 g
UNO Board	25 g
Pont H	23 g
Stepper Motor	163 g
pilote moteur	7 g
Potentiometer	9 g
Servo Motor	12 g
Battery Connector	4 g
Module d'alimentation	11 g
boussole	3 g
Motor Driver	27 g
LCD Display	33 g
Speed Sensor	19 g
GPS	13 g
Connection Cables	30 g
Gear	42 g
Propeller	36 g
Stern tube	101 g
Coupling	12 g
Drive shaft	47 g
Internal support	150 g
Hull	1300 g
Total	2224 g 2.2 kg

4. Displaced Volume at Equilibrium :

At equilibrium, the boat's weight (p) equals the buoyant force : P=Fa therefore , P=p.V.g
With =1000 kg.m3 and g=9.81 m2.s
So,
V=P.g / p AN : V= 21.91000 / 9,81 = 0.0022 m3

5. Immersion Height of the hull (h)

For a this boat with width and length , the displaced volume is given by : V=1/2(L). B. h
With l=16.98 cm and L=70 cm
*Knowing that l=B h=2vB / L ; AN : h=2 (0,0022)0,1698 / 0,70 = 0,037 m

6. Center of Buoyancy (CB)

It is the center of volume (and also the center of gravity of the fluid) displaced by the hull of a ship. It is the point of application of the Archimedes thrust which makes the ship float.

- Xc=0,7 / 2=0,35 m
- Yc=0,1698 / 2=0.0849 m
- Zc=0.037 / 2=0.0185 m

⇒ CB = {0,35 m; 0,0849 m; 0,0185 m}

III/ STABILITY BOAT

Stability refers to the boat's ability to resist inclinations. It is characterized by the center of gravity (G), the center of buoyancy (C), and the metacenter (M).

l. Definitions of terms :

- G (Center of Gravity) : Where all the weight of the boat acts downwards. It depends on how your components are placed inside the boat.
- B (Center of Buoyancy): Where the upward force of water (Archimedes' force) acts. It's the center of the submerged part of your boat.
- M (Méta Center) : The point where the new vertical buoyant force acts when the boat tilts. Used to assess stability.

Now, if your boat tilts (leans to the side), these two forces try to bring it back or flip it over. That's where the other points come in.

- BG (Distance from B to G) : Vertical distance between center of buoyancy and center of gravity.
- BM (Distance from B to M) : Geometry-based value that shows how fast the center of buoyancy moves when the boat tilts.
- GM (Distance from G to M) : The most important value: tells you if the boat is stable or not.

2. GM Hypothesis and Calculations :

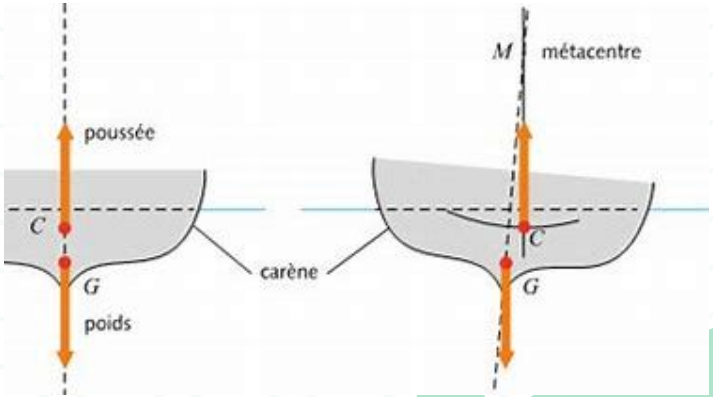
- If GM > 0 ⟶ Your boat is stable (it rights itself after tilting).
- If GM < 0 ⟶ Your boat is unstable (it might tip over).

The bigger GM is the more stable, but if it's too big, the boat can roll too fast (jerky movement).
It is given by : GM=BM-BG.

➔ Calculate : BG=G-B
B=h / 2 ; AN : B=0,037 / 2=0,0185 m so estimate G=0,045 m ;
AN : BG=0,045-0,0185=0,0265 m 2,65 cm

➔ Calculate : BM=IV
I=L . l^3 / 12 ; AN : I=0,7 . (0,1698)^3 / 12=0,0002858 m4
AN : BM=0,00028580,0022=0,13 m 13 cm

Finally, AN : GM=0,13-0,0265=0,1035 m ~10,35 m



3. Righting Moment (RM) :

This is also called the righting torque, it is equal to the product of the boat's displacement by the horizontal distance between the verticals passing through the gravity and the hull Center. It is given by : RM=P . GZ

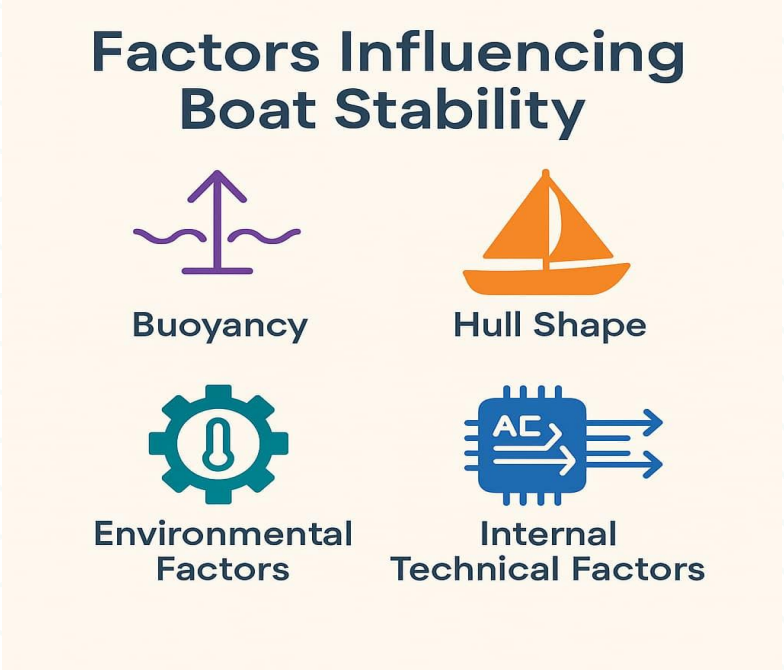
➔ Calculate : GZ=GM . sin()

For an angle of inclination =10; AN : GZ=10,35 . sin(0,1745)=0,0179 m

Finally, AN : RM=21,9 . 0,01797 = **0,393 Nm**

4. Influencing factors :

There are many factors that can influence not only the balance but also the flotation of our mini boat on the water, among which:



1. Internal physical and technical factors :

These are the design and internal characteristics that influence the stability, buoyancy, and dynamic behavior of the mini-boat.

Factors	Effects
Hull shape (V-shaped)	Offers better penetration into the water, reduces drag, and improves directional stability. It also provides better handling in slightly disturbed waters.
Mass distribution	Poor distribution raises the center of gravity(G) decreases GM; and may cause instability.
Material used (ABS)	Lightweight and resistant, ABS enhances buoyancy while keeping the structure rigid.
Positioning of heavy components (battery, motor)	If placed too high or off-center, they raise the risk of capsizing. Ideally, they should be low and close to the longitudinal center.
Submerged surface and hull profile	A larger surface increases water resistance (drag).
Hydrodynamic forces	Water flow around the hull creates drag and lateral forces. A streamlined V-shaped hull helps minimize these effects.
Aerodynamic forces	Wind can apply lateral force to the boat, especially if external structures (e.g. GPS, sensors) stick out. This can create a capsizing moment.

2. External Environmental factors :

Factors	Effects
Waves/water motion	Can cause pitching (forward-backward) or rolling (side-to-side) may threaten stability.
Wind	May push the boat or tilt it if it acts as an elevated structure.
Water currents	Can disrupt trajectory or push the boat off course.
Température	Affects materials expansion and water density.
Type of water (fresh/salt)	Saltwater is more dense better buoyancy. In freshwater, the boat sits slightly lower.
Floating debris or obstacles	May cause impact or disturbances requires effective navigation.

The behavior of our mini-boat is influenced by a combination of internal factors—such as the V-shaped hull, the use of ABS material, the placement of internal components, and hydrodynamic/aerodynamic forces—and external factors, including wind, current, and water conditions. Our design choices aim to optimize these elements to ensure buoyancy, stability, and performance, even under moderately challenging environmental conditions.

5. Environmental Impact :

Here, we are basing ourselves on the life Cycle Analysis (LCA) method and the carbon footprint of our mini-boat.

1. Life Cycle Analysis :



I. Life Cycle Analysis :

Life Cycle Assessment (LCA) is a method used to evaluate the environmental impacts of a product from the extraction of raw materials to its end of life. For our mini-boat, this analysis allows us to identify the sources of pollution, energy consumption, and areas for improvement.

- ### Goal of the LCA :

The main goal of the (LCA) of our mini-boat is to evaluate its environmental impacts at every stage of its life cycle – from raw material extraction, manufacturing, assembly, usage, maintenance, to end-of-life disposal. It help us, to :

- Identify the main sources of CO_2 emissions and pollution,
- Make informed design and material choices,
- Optimize the use of resources (energy, materials),

- Promote sustainability in engineering education,
- Reduce the ecological footprint of the prototype.

- ## Functional Unit :

For our project, the functional unit is:
"One functional prototype of a 70 cm mini-boat, designed for educational navigation and stability testing, with a lifespan of 1 academic year and reusable components."
This unit allows us to measure environmental impacts (like CO_2 emissions) per unit of educational use, ensuring consistency and relevance in our evaluation.

- ## Life Cycle Stages :

LIFE CYCLE ANALYSIS

BOAT PROJECT

Stage 1

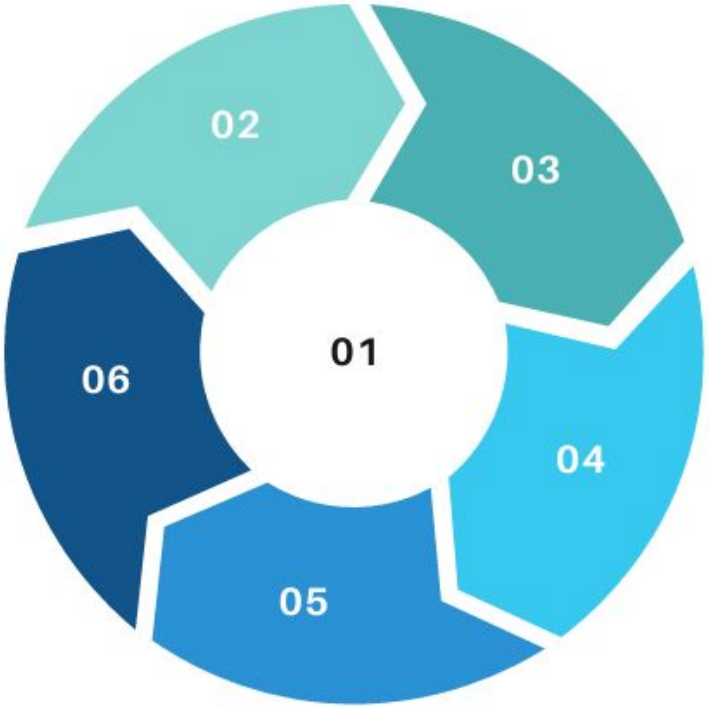
Raw Material Extraction an Manufacturing

Stage 2

Transport

Stage 3

Assemby



Stage 4

Usage

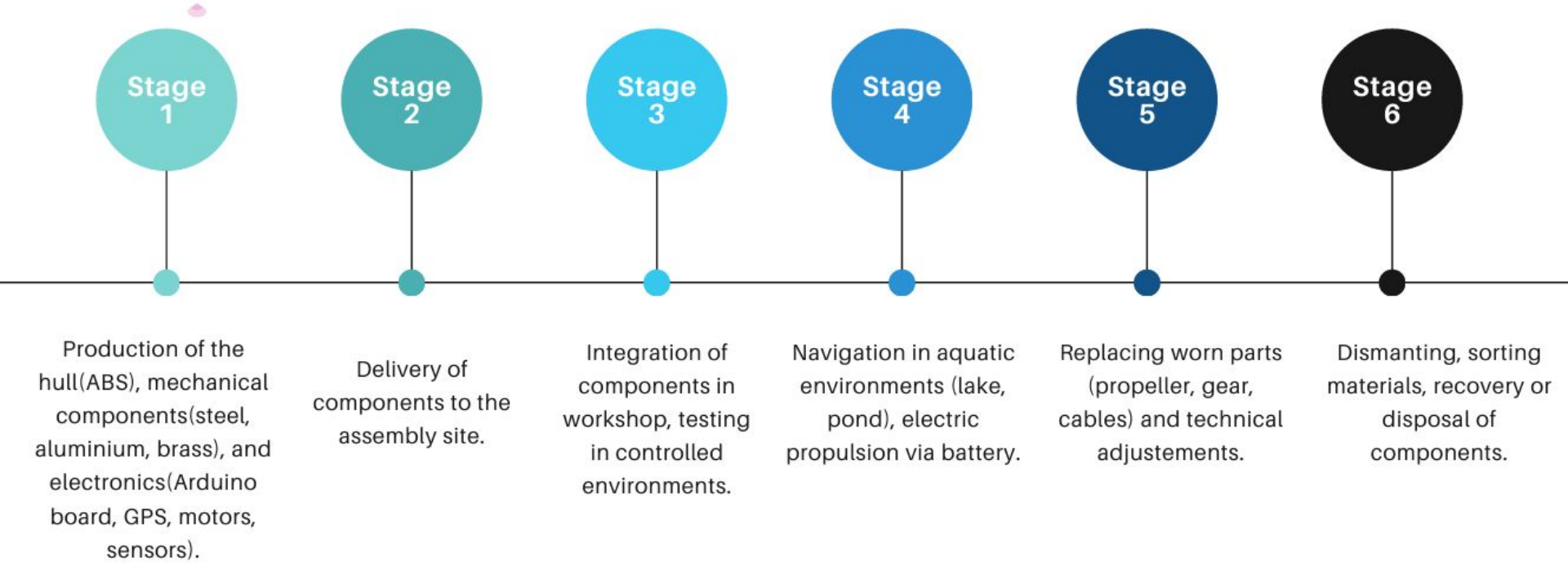
Stage 5

Maintenance

Stage 6

End of life

- Description



- Environment Impacts

Stage 1

CO2 emissions from industrial processes:
1.7kg CO2/kg of aluminum;
7.1kg CO2/kg of steel

Stage 2

On average, transport accounts for 6-10% of a technical product's total carbon footprint.

Stage 3

Low electricity consumption (0.2kwh); around 0.1kg Co2

Stage 4

A 7.4V-1000mAh Li-ion battery indirectly emits 0.4kg CO2 per charge cycle

Stage 5

Low-frequency technical waste production

Stage 6

Without proper sorting; 80% of plastic components end up in landfills. Improper disposal of electronics = hazardous pllution (lead or lithium)

1. Estimated Carbon Footprint of the mini-boat :

The carbon footprint represents the total amount of greenhouse gases (mainly CO_2) emitted throughout the life cycle of our mini-boat.
For our prototype, we estimate a total emission of approximately 19 to 20 kg of CO_2.

Components	Quality	Estimated CO2 emissions
ABS Hull (2,3 kg)	2,3 kg	3,9 kg CO2
Aluminium (coupling)	0,2 kg	1,64 kg CO2
Steel (shaft)	0,3 kg	2,1 kg CO2
Brass (stern tube)	0,1 kg	0,6 kg CO2
Electronics (Arduino, GPS, motors)	0,5 kg	10 kg CO2
Rechargeable battery (Li-ion)	0,2 kg	1,2 kg CO2 (average estimate)
Total estimated	3,6 kg of materials	1,4 kg CO2 across the full life cycle

2. Co-Design practices Implemented ;

To reduce our impact, we implemented several responsible choices:

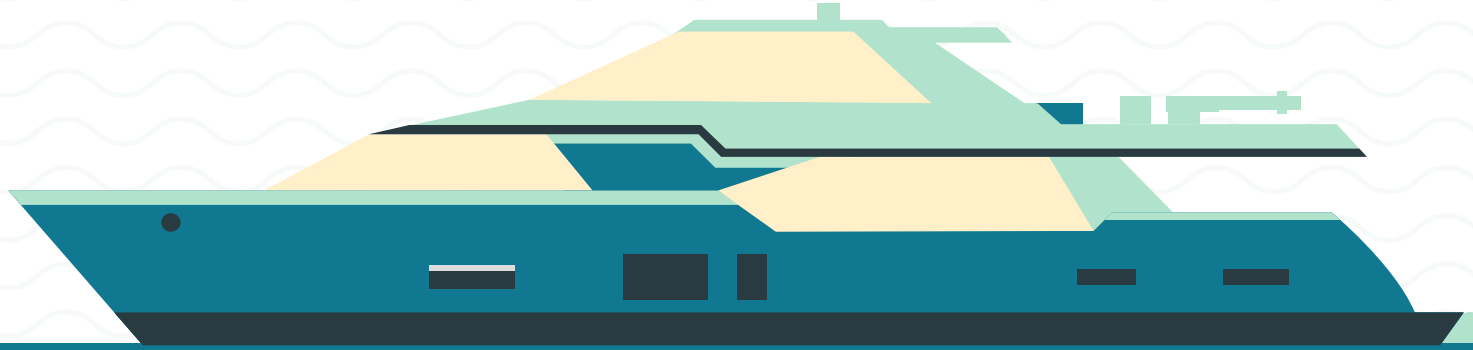
♻️ Reusable components (hull, battery, electronics) for future projects.
🔌 Electric propulsion → zero direct emissions during operation.
📦 Easy disassembly for better end-of-life material sorting.
🧰 Preventive maintenance to extend the boat's lifespan.
❌ Reduced plastic packaging when ordering components.

Although small and educational, our mini-boat generates an estimated 19 to 20 kg of CO_2 throughout its life cycle. This is equivalent to driving approximately 120 km by car (source: ADEME, 2023). Through thoughtful design choices—such as using ABS, reusing components, and relying on rechargeable batteries—we managed to significantly lower the environmental impact, while also raising awareness of eco-design practices in the engineering process.

05

EEE Domain

ABO WILLIAM JUNIOR



SUMMARY OF THE DOMAIN

- Objective and explanation of contents
- Work principle and role of each component
- Controlling the speed and rotation torque of the motor
- Command, Information chain and circuit supply chain
- Digital functional blocks synthesis diagram

  representation circuit
- Simulation and physical presentation of the model

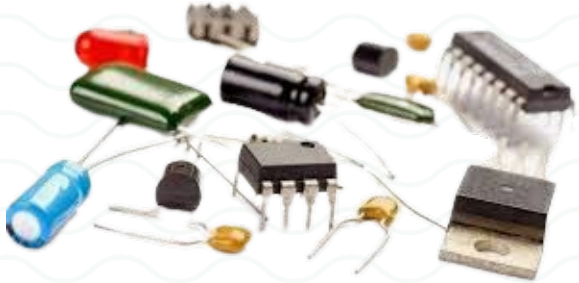


Objectives and explanation of contents

The electrical and electronic engineering (EEE) section of a boat project typically covers the design, installation, and integration of all the electrical systems and electronic components on the vessel.

During the first semester, as part of the electrotechnical boat project, electronic and automatic (EEE), the deliverable was to Break down the motor control system into functional blocks, design and simulate the Analogic functional blocks model.

Now during this second semester we are going to focus on the Breakdown of the digital part of the motor control into functional blocks, design and simulate the blocks using suitable software and test them on suitable hardware.



Work principle and role of each components

1- DC MOTOR

A DC motor is an electrical motor that uses direct current (DC) to produce mechanical force.The most common types rely on magnetic forces produced by currents in the coils. Nearly all types of DC motors have some internal mechanism, either electromechanical or electronic, to periodically change the direction of current in part of the motor.



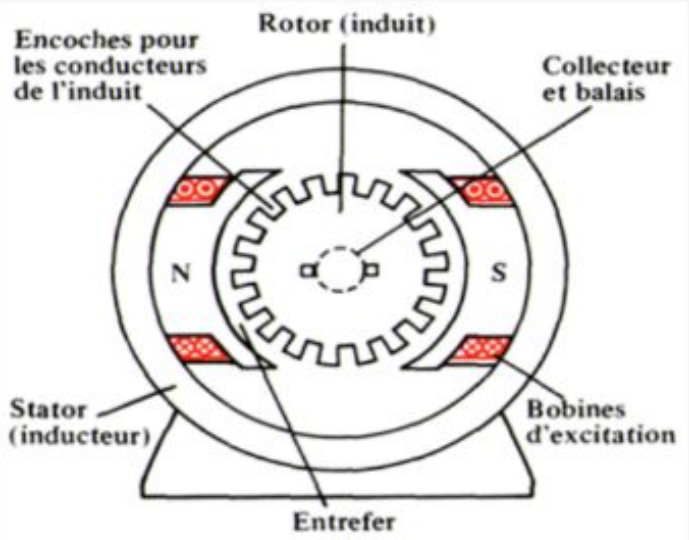


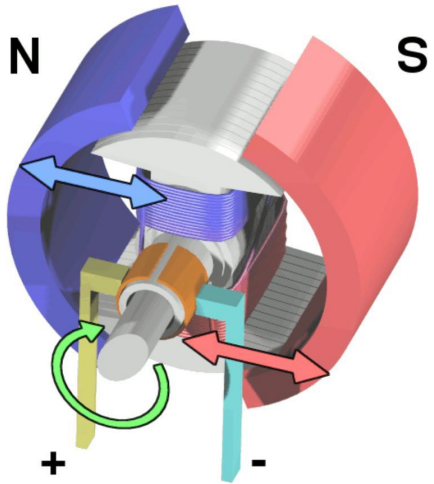
ATTRIBUT	VALUE
Supply voltage	12 Vcc
Power output	14W
Output speed	4289 tr/min
Shaft diameter	6 mm
Maximum output torque	350 gcm
Length	133.5mm
Dimensions	(Ø) 37 mm
Current	11.5 A
Core material	Steel Core
Weight	157g



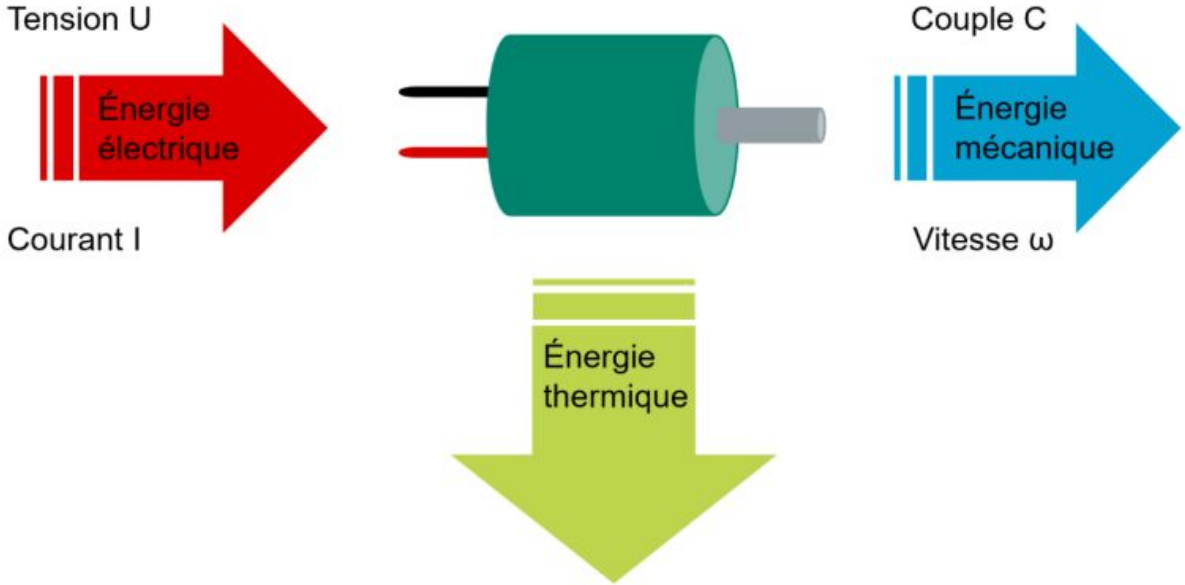
- **Principle of Dc motor :**

A DC motor is made up of a magnetic circuit featuring a fixed part named stator, a rotative part named rotor and air-gap between both parts:The inductor (stator) creates fields which rotate by winding.An induced electrical circuit (rotor) is affected by this magnetic field. The collector and brushes provide access to the rotor circuit.





TECHNICAL NOTIONS



Electrical power (current/voltage) is converted into mechanical power (speed/torque) by the electric motor. The losses associated with this transformation are losses due to friction (mechanical) and heat dissipation (Joule losses) in the winding. For brushless motors, Joule losses are combined with friction losses.

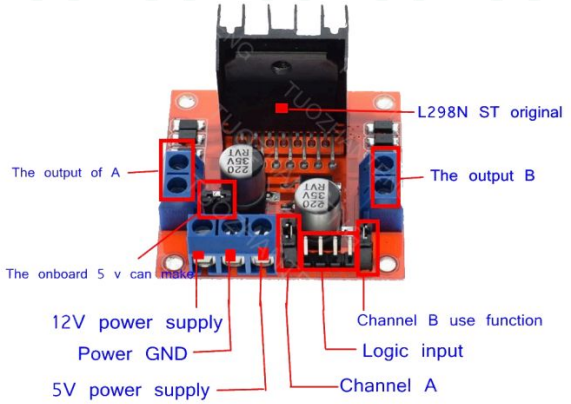
2- Arduino microcontroller board

Arduino senses the environment by receiving input from many sensors, and affects its surroundings by controlling lights, motors, and other actuators.

3- DC Motor Driver(H-Bridge)

The Dual DC Motor Driver 4A V3 allows two DC motors to be driven independently, controlling both speed and direction. The H-bridge receives the PWM signal as input and regulates the quantity of energy supplied to the motor.





4- Power supply

It's essential to have an adequate source of energy to provide the voltage and current required for the operation of the H-bridge and the motor. This source can be a battery, a regulated power supply or any other source suitable for energy.

5- Button

We've included a button to change the direction of rotation of our DC motor

6- Potentiometer

A user-adjustable 3 terminal variable resistor that can be used as a voltage divider. The first one will be used as a dimmer for the LCD screen, the second one will be used to set the boat direction.

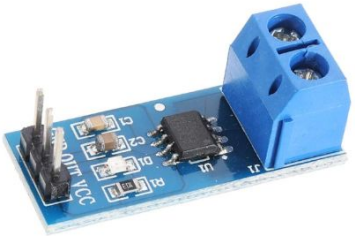


7-Current and Speed Sensors

- Speed sensor measures the rotational speed of the motor by detecting the movement of the motor shaft and generating a corresponding output signal, allowing the speed to be calculated.



- Current sensor using a sensor effect Hall, this device measures the current consumed by the motor. This information is essential for monitoring power consumption and preventing overcurrent.



Command, Information and supply chain

1. Control System Breakdown

- **External Data:**

This refers to instructions sent by a user or a higher-level system: desired speed, direction, operating mode, etc. It can come from a remote control, a setpoint sensor, or a human-machine interface (HMI).

- **Conversion:**

This block transforms physical (analog) data from sensors (e.g., voltage, current, speed) into digital signals that can be used by the control board. It is an analog-to-digital (A/D) interface. It also supplies the sensors with current (VDC).

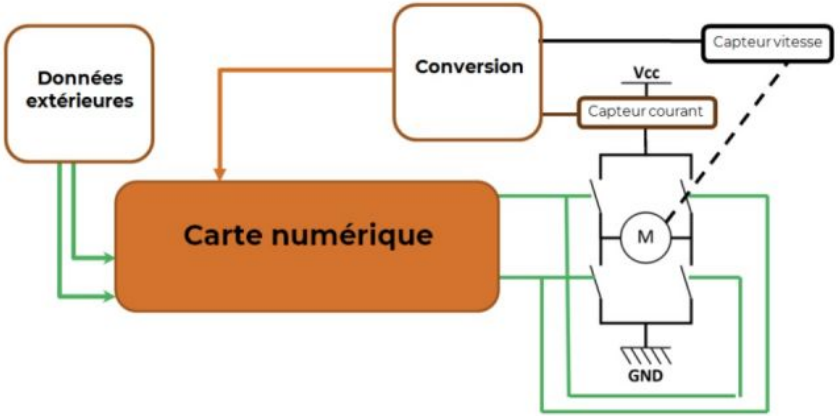
- **Digital board:**

This is the heart of the control system. It can be a microcontroller, an FPGA, or an Arduino. It executes a control program that:

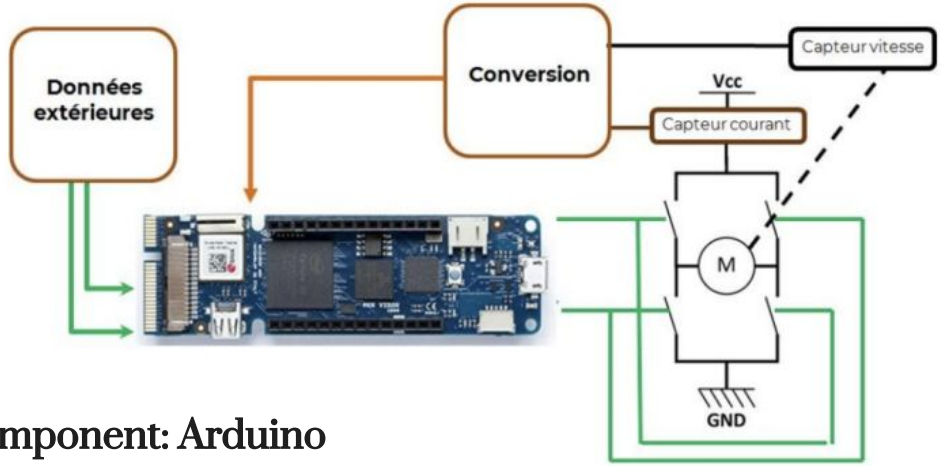
- Reads data from the sensors via conversion.
- Compares the measurements with the setpoints.
- Calculates a command (often PWM) to adjust the motor.

Overall system operation:

1. The user (or another machine) sends a command via external data.
2. The digital board receives this data, then reads the actual state of the motor using the sensors.
3. The conversion transforms the physical signals from the sensors into usable digital values.
4. The board compares the command and the actual state and calculates a corrective command.
5. This command is sent to the motor to correct its speed or direction.
6. The loop repeats in real time: it is a closed-loop system.



2. Automated system (with Arduino)



3. Central component: Arduino

Arduino is an open-source microcontroller board widely used in educational and semi-professional embedded systems.

- Reads values from sensors (via analog or digital pins).
- Executes a control program (e.g., PID).
- Generates a PWM signal to the motor.

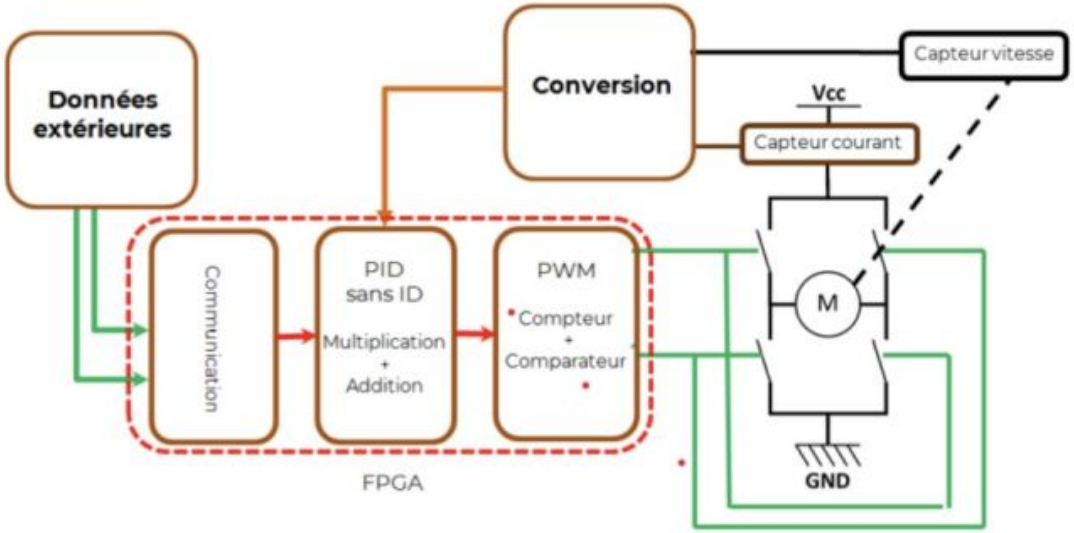
Can also communicate with a PC, radio, or another Arduino.

3. FPGA-based control loop with motor and current sensor

- FPGA (Field Programmable Gate Array): Heart of the processing system.

- Communication: Receives and exchanges data with the environment (Arduino, radio, etc.).
- ID-less PID: Proportional + Derivative (PD) controller. It adjusts the action on the motor based on the difference between the desired and actual speed, but without integration.
- Multiplication / Addition: Internal calculations within the PID to obtain the correction command.
- PWM (Pulse Width Modulation): Generates a square waveform whose duty cycle controls the motor's power.
- Counter: Generates a time base.
- Comparator: Compares the setpoint with the time base to create the PWM signal.

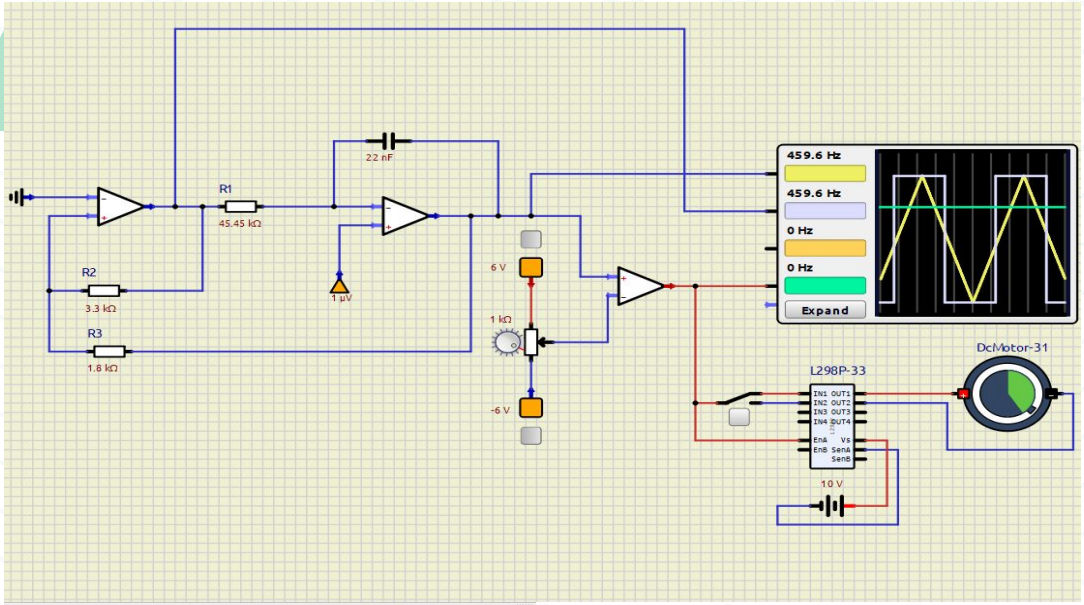
- M (Motor): Executes the command (speed or direction).

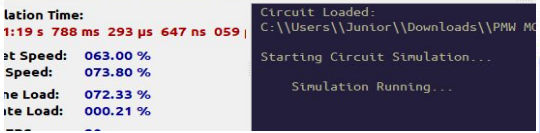


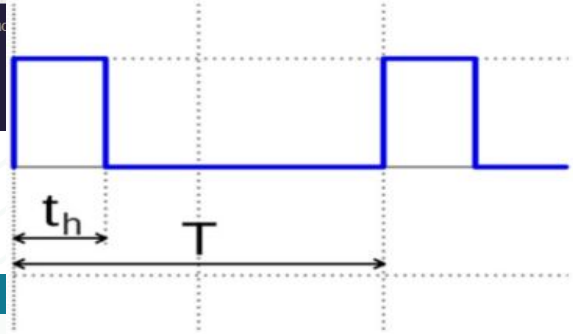
Overall operation:

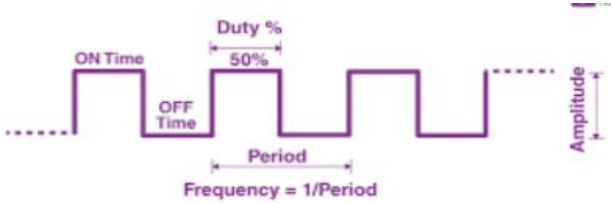
1. The system receives a speed command.
2. The current sensor verifies the force applied by the motor.
3. The FPGA dynamically adjusts the PWM based on the measurement using its PD controller.
4. The motor reacts to the PWM, changing its speed.

How the PWM Signal is being determined



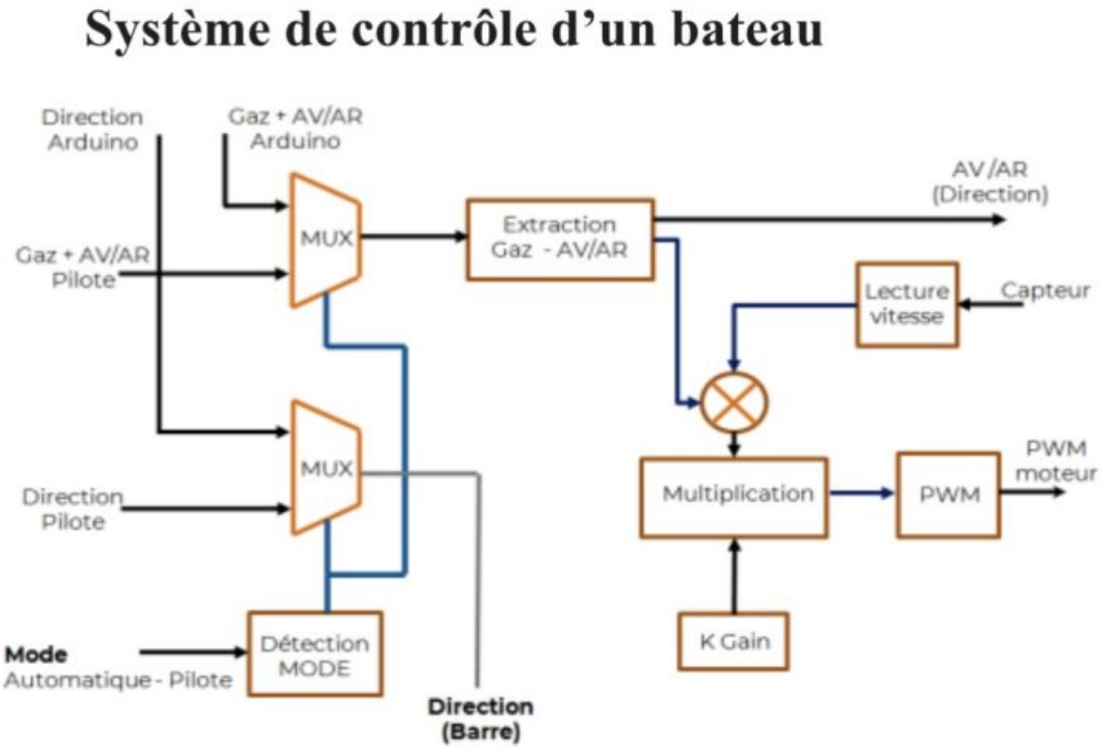






Summary diagram and simulation

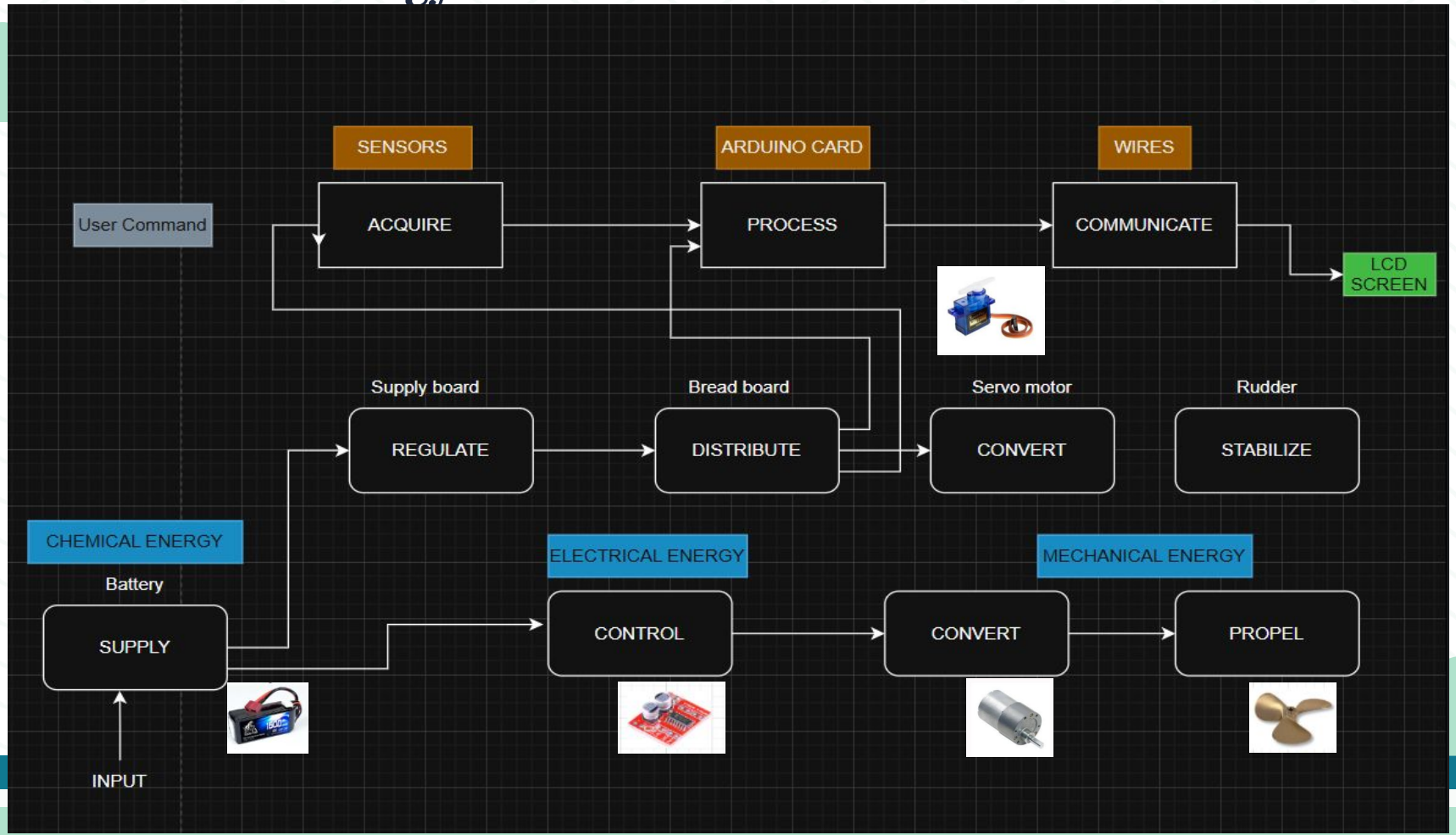
4. Boat directional control system and engine



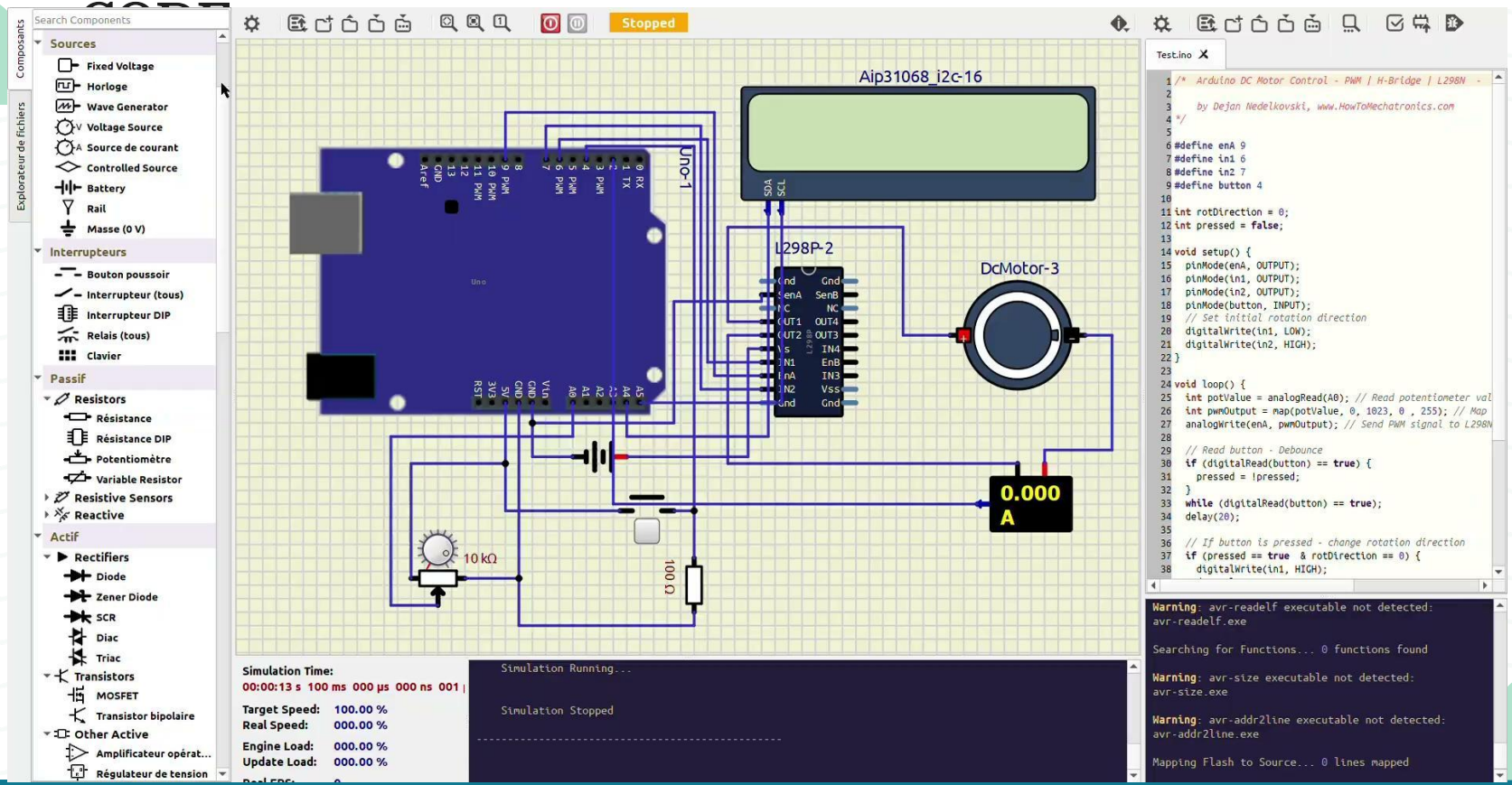
Overall operations :

1. The mode (**manual or automatic**) is detected, and the **MUX routes** the signals from either the **Arduino** or the **pilot** accordingly.
2. The **steering and speed signals** are analyzed.
3. The **actual speed** is measured using a sensor.
4. The **error** between the setpoint (desired speed) and the actual speed is **multiplied by K Gain**.
5. The result is sent to the **PWM module**, which adjusts the motor accordingly.
6. The **motor responds** (moves forward, reverses, or changes direction).

Information and Energy chain



FINAL SIMULATION AND



MeasureCurrent.ino

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>

LiquidCrystal_I2C lcd(0x3F, 16, 2); // Adresse 0x3F, écran 16 colonnes, 2 lignes

#define enA 9
#define POTENTIOMETER_PIN A0
#define CURRENT_SENSOR_PIN A1
#define IN1 6     // Broche PWM pour IN1
#define IN2 7     // Broche PWM pour IN2
#define SPEED_SENSOR_PIN 2
#define DIRECTION_BUTTON_PIN 4  // Bouton pour changer de sens

const float ACS712_SENSITIVITY = 0.185;
const int ACS712_OFFSET = 512;

volatile unsigned int pulseCount = 0;
float motorRPM = 0.0;

bool motorDirection = true;           // true = avant, false = arrière
bool lastButtonState = HIGH;
```

MeasureCurrent.ino

```
void setup() {
  pinMode(enA, OUTPUT);
  pinMode(IN1, OUTPUT);
  pinMode(IN2, OUTPUT);
  pinMode(SPEED_SENSOR_PIN, INPUT_PULLUP);
  pinMode(DIRECTION_BUTTON_PIN, INPUT_PULLUP);

  attachInterrupt(digitalPinToInterrupt(SPEED_SENSOR_PIN), countPulse, FALLING);

  lcd.init();
  lcd.backlight();
  lcd.print("Controle moteur");
  delay(2000);
  lcd.clear();

  digitalWrite(IN1, LOW);
  digitalWrite(IN2, LOW);
}

void loop() {
  // Lecture et gestion du bouton (anti-rebond)
```

Output

MeasureCurrent.ino

```
67    // Application PWM sur IN1 ou IN2 selon direction
68    if (motorDirection) {
69      analogWrite(IN1, motorSpeed);
70      digitalWrite(IN2, LOW);
71    } else {
72      digitalWrite(IN1, LOW);
73      analogWrite(IN2, motorSpeed);
74    }
75
76    // Calcul vitesse moteur et courant
77    motorRPM = calculateRPM();
78    float current = readCurrent();
79
80    // Affichage LCD
81    lcd.setCursor(0, 0);
82    lcd.print("Speed:");
83    lcd.print(motorSpeed);
84    lcd.print("PWM ");
85
86    lcd.setCursor(0, 1);
87    lcd.print("RPM:");
```

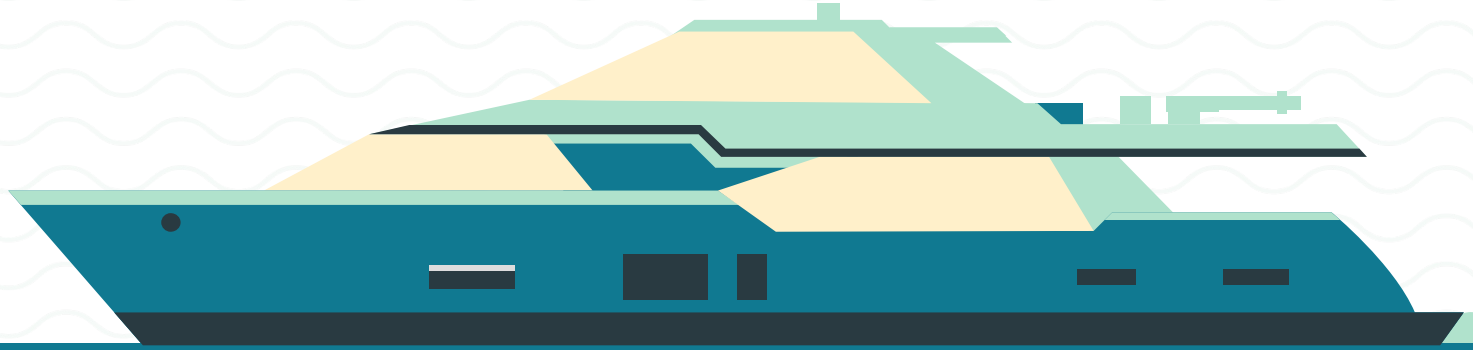
MeasureCurrent.ino

```
86      lcd.setCursor(0, 1);
87      lcd.print("RPM:");
88      lcd.print(motorRPM, 1);
89      lcd.print(" I:");
90      lcd.print(current, 2);
91      lcd.print("A ");
92
93      delay(500);
94    }
95
96    void countPulse() {
97      pulseCount++;
98    }
99
100   float calculateRPM() {
101     static unsigned long lastTime = 0;
102     unsigned long currentTime = millis();
103     unsigned int pulses;
104
105     noInterrupts();
106     pulses = pulseCount;
```

06

MIA Domain

MAKOSSO JEAN CLAUDE MIGNON



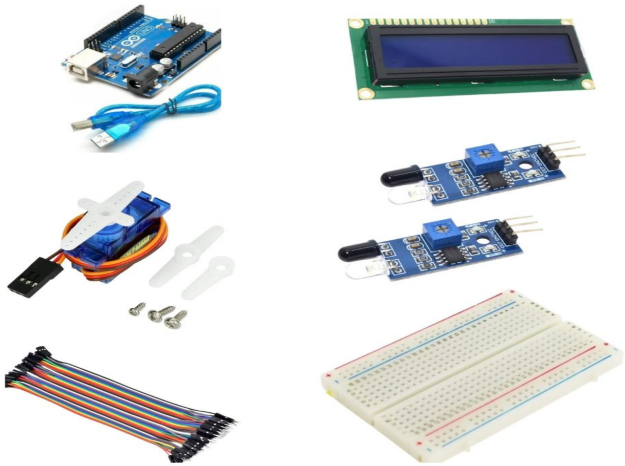
SUMMARY OF THE DOMAIN

- Objectives of the Domain
- Components used
- Working Principle
- Wiring diagram
- Arduino code
- Correction system
- Differential equation



Objectives of the domain

Design a simple autopilot system using Arduino. Control the boat's direction automatically using a servo motor and a magnetometer.



Components used

- Arduino UNO
- potentiometer
- Servo Motor (to simulate rudder)
- I2C LCD 16x2 display
- Magnetometer(GY-271-HMC5883L)
- Jumper wires and beadboard

Working principle

- The potentiometer sets the target angle
- The magnetometer reads the actual heading of the boat
- Arduino compares both values and calculates the angular error
- The servo motor adjusts automatically to minimize the error
- LCD displays target angle and heading



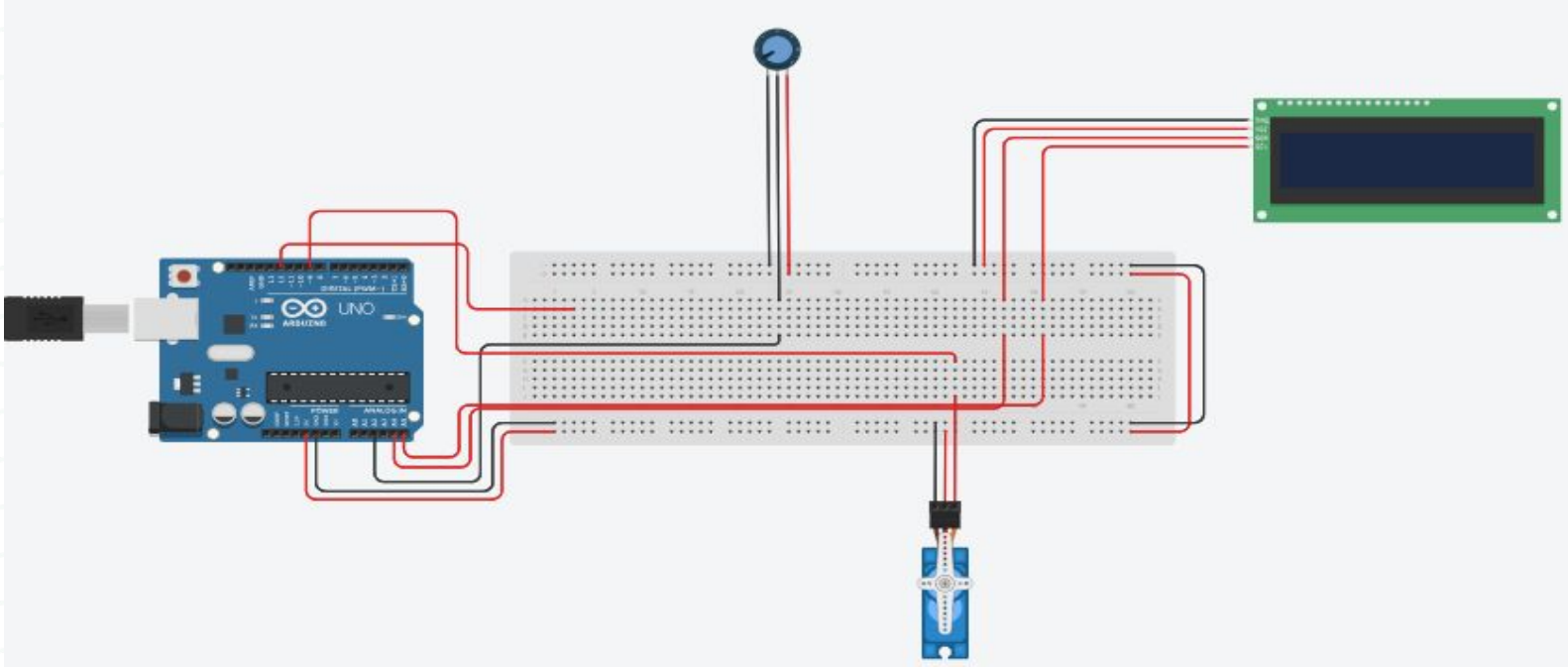








Wiring Diagram



Arduino code

```
#include <Wire.h>
#include <Servo.h>
#include <HMC5883L.h>

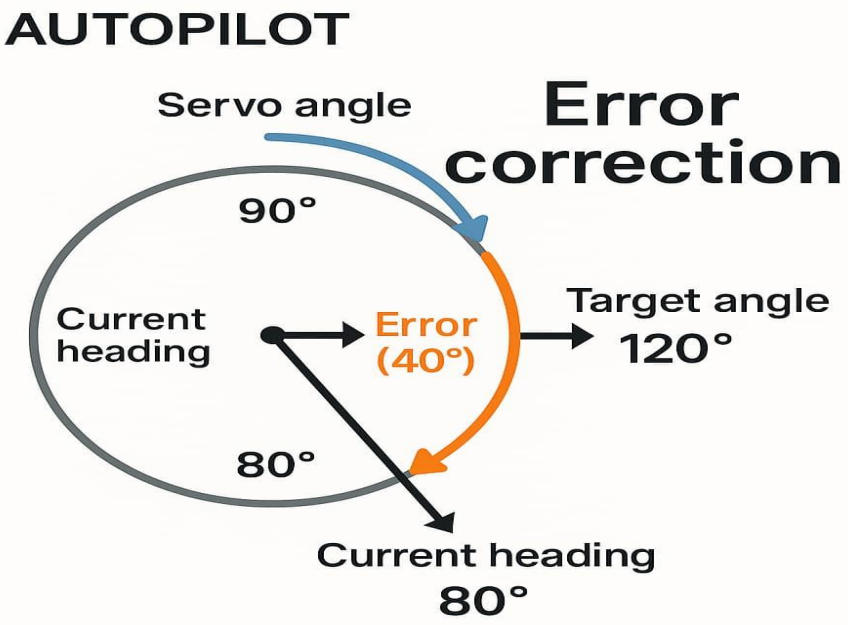
Servo rudder;
HMC5883L compass;

void loop() {
  int target = map(analogRead(A2),
0, 1023, 0, 180);
  Vector dir =
compass.readNormalize();
  int heading = atan2(dir.YAxis,
dir.XAxis) * 180 / PI;
  if (heading < 0) heading += 360;

  int error = target - heading;
  if (error > 180) error -= 360;
  if (error < -180) error += 360;

  rudder.write(constrain(90 + error,
0, 180));
}
```

Correction system



1. error = targetAngle - currentHeading

targetAngle = 120
currentHeading = 80
error = 120 - 80 = 40

2. servoAngle = 90 + error
If error = 120 → servoAngle = 90 (he stays right)

if error > 120 → he turns right

Si error < 120 → he turns left

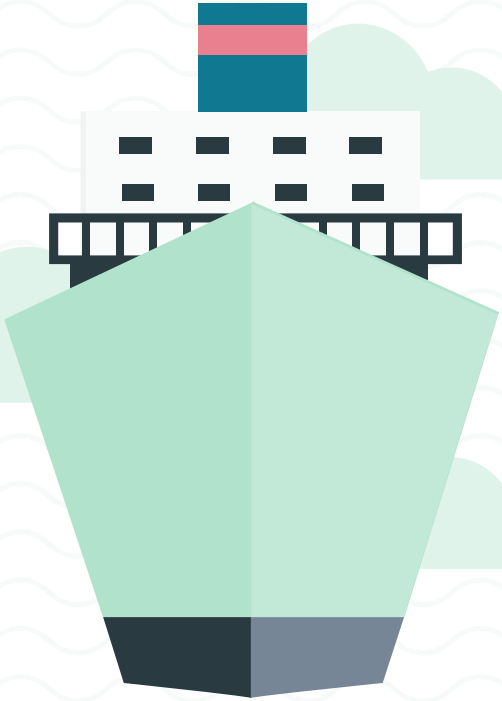
Differential equation

$$(T.d\Theta/dt)+\Theta(t)=\Theta_{target}(t)$$

$$0.5 \times d\Theta/dt = 120 - 80 = 40$$

$$d\Theta/dt = 40/0.5 = 80°/s$$

Element	Value	Meaning
Error (angle)	40°	The distance between heading and target
T (time constant)	0.5s	System response time
$d\Theta/dt$	80°/s	Servo correction speed




A picture is worth a thousand words

Thanks!

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